

Copyright Notice

These slides are distributed under the Creative Commons License.

[DeepLearning.AI](#) makes these slides available for educational purposes. You may not use or distribute these slides for commercial purposes. You may make copies of these slides and use or distribute them for educational purposes as long as you cite [DeepLearning.AI](#) as the source of the slides.

For the rest of the details of the license, see <https://creativecommons.org/licenses/by-sa/2.0/legalcode>

Applied Statistics for Data Analytics

Module 1: Foundational
statistical techniques





Applied Statistics for Data Analytics

Welcome to this course



Applied Statistics for Data Analytics

Generative AI in this course

In this course, you'll...



Learn how to use LLMs to:

- Troubleshoot spreadsheet errors
- Create custom conditional formatting
- Design and run simulations
- Interpret inferential statistics
- Run statistical analysis for you



Explore LLMs' limitations, learning when to:

- Select the LLM for a particular task
- Use something else, like a spreadsheet

LLMs in this course



Demonstrates the most up-to-date capabilities as of 2025



Evergreen principles:

- How to **think about** and **use** generative AI regardless of product



Develop a mindset of iteration and skepticism

- New models and features constantly released

Changes you should expect

- More advanced and specialized features
- Cheaper tools
- Faster tools
- Higher quality outputs overall

LLM options

✗ You **won't** need to purchase any additional products

✗ This course **does not** recommended any single tool

✓ You'll develop confidence experimenting and selecting tools

✓ You'll see several tools throughout the modules

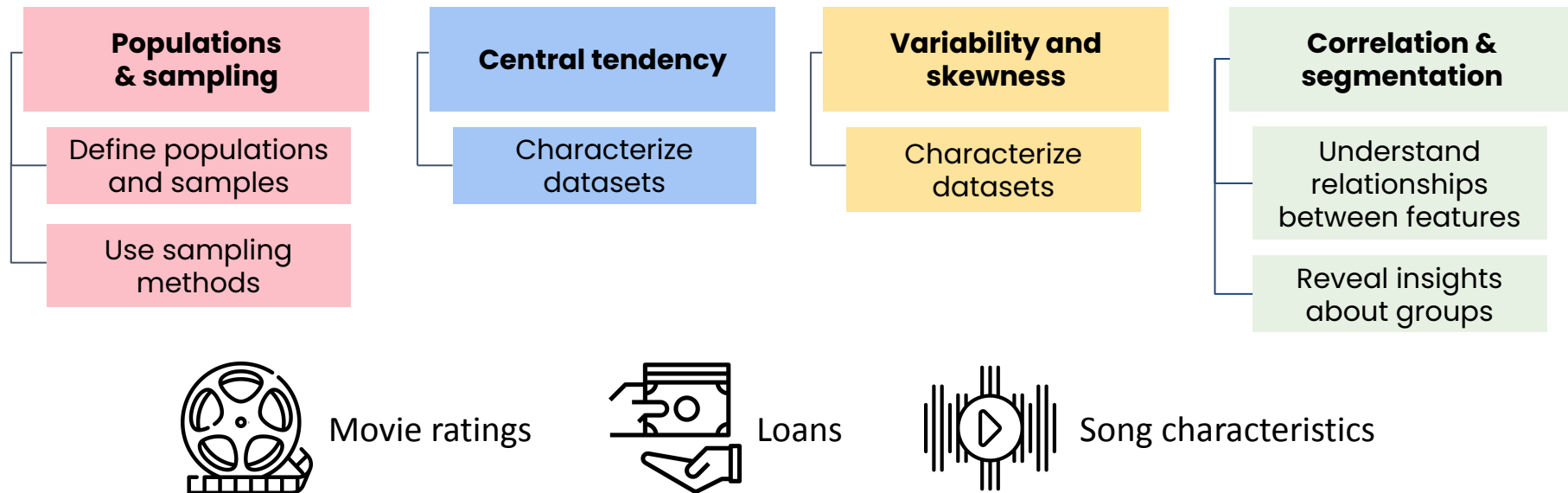
✓ You'll learn core principles to work with LLMs: free and paid, now and in the future



Applied Statistics for Data Analytics

Module 1 introduction

Module 1 outline





Foundational statistical techniques

Populations and sampling

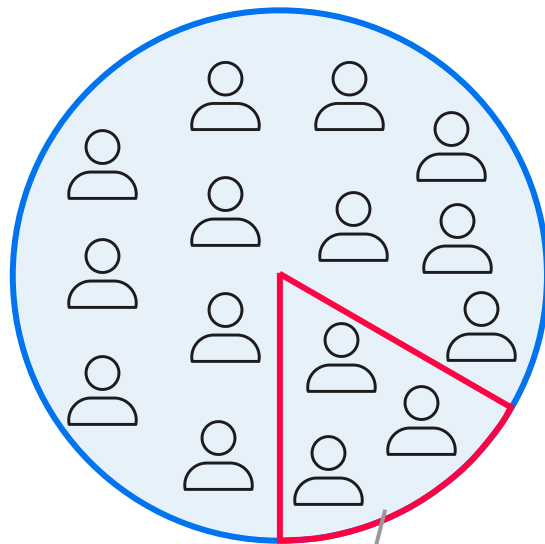
Population & sample

Population

The set of all individuals or observations you have a hypothesis about

Sample

A subset of that population that has been chosen for analysis



Scenario: population & sample



Hypothesis: Majority of coworkers support signed e-card.



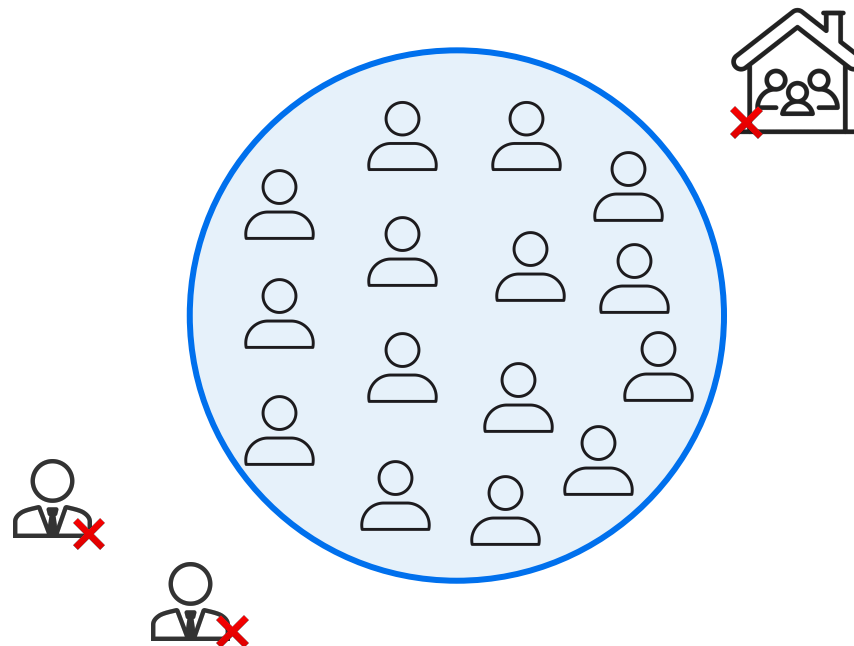
Population: Everyone working at your company



Not included:

- Company's clients
- Your family

Population



Scenario: population & sample



Goal: Figure out who's in favor of this birthday card



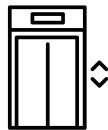
Tedious to ask everyone's opinion



Collect a sample:



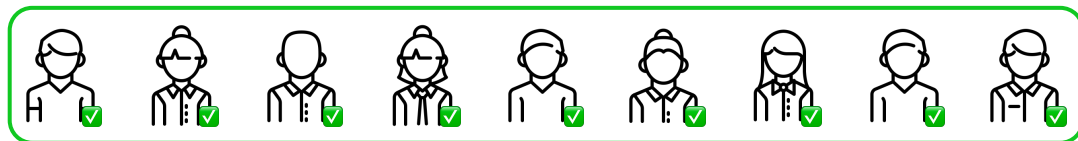
Poll



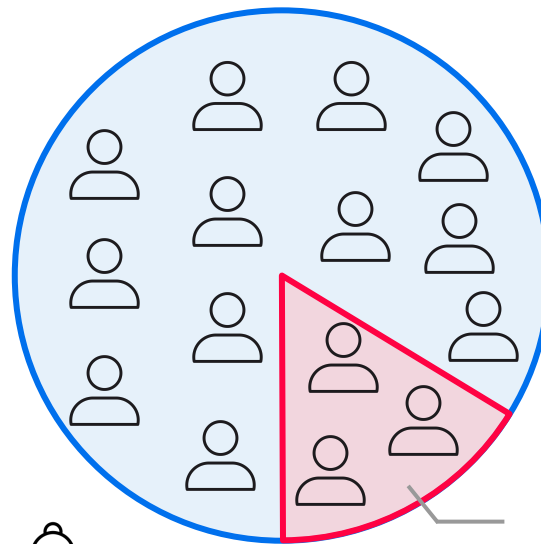
Chat



Survey



Population



Sample

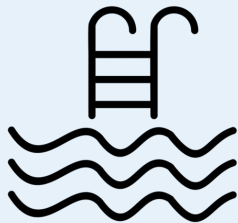
A subset of the population
chosen for analysis

Population examples

A population doesn't
have to refer to people

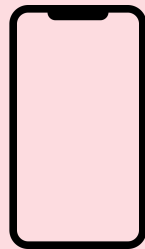
Population:

Hypothesis:



Water in a pool

Water quality allows
for swimming



All iPhones
manufactured
in a facility

<1% have defects



Every bald eagle
in the US

Nesting behavior
varies by region

Why this matters

You won't typically have access to entire population.

Accessible 

 50 coworkers

Not accessible 

 200 million
Netflix users

1. **Knowledge**



Population might be unknown

2. **Practicality**



Too much time



Too much money



Might not be able to access its entirety

3. **Ethics**



Testing could cause harm

The power of statistics

A relatively **small sample** can accurately characterize the population.

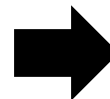
Example: Test the water quality of a swimming pool

- ✗ Don't need to collect every drop of water
- ✓ 500 milliliters is more than enough

Population



Sample



Example: 1997 Report “Where America Stands”

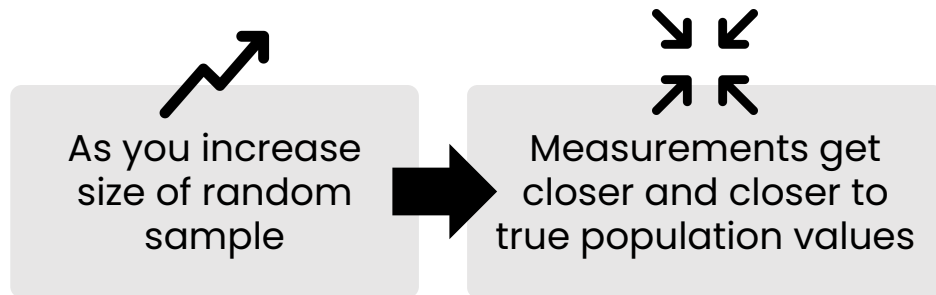
- Gallup used **surveys** to estimate what Americans thought about:

 Gay rights  Smoking  President

Population

187,000,000
Americans

The Law of Large Numbers



Sample

1,000
surveyed

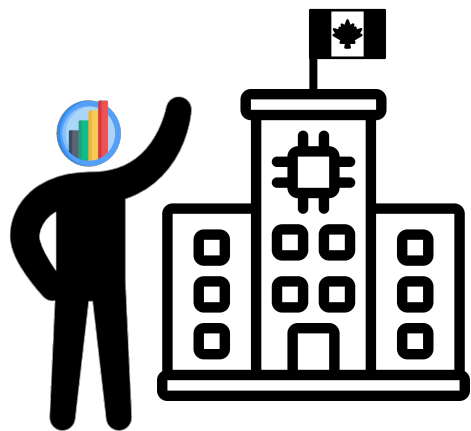
Accurate within a few percentage points



Foundational statistical techniques

Identifying the population

Scenario



You
Data Analyst



Problem: "Can you figure out why people aren't signing up?"



Observation: "People who aren't signing up" is pretty vague.



Potential populations of interest:



People who used to use your AI service but canceled

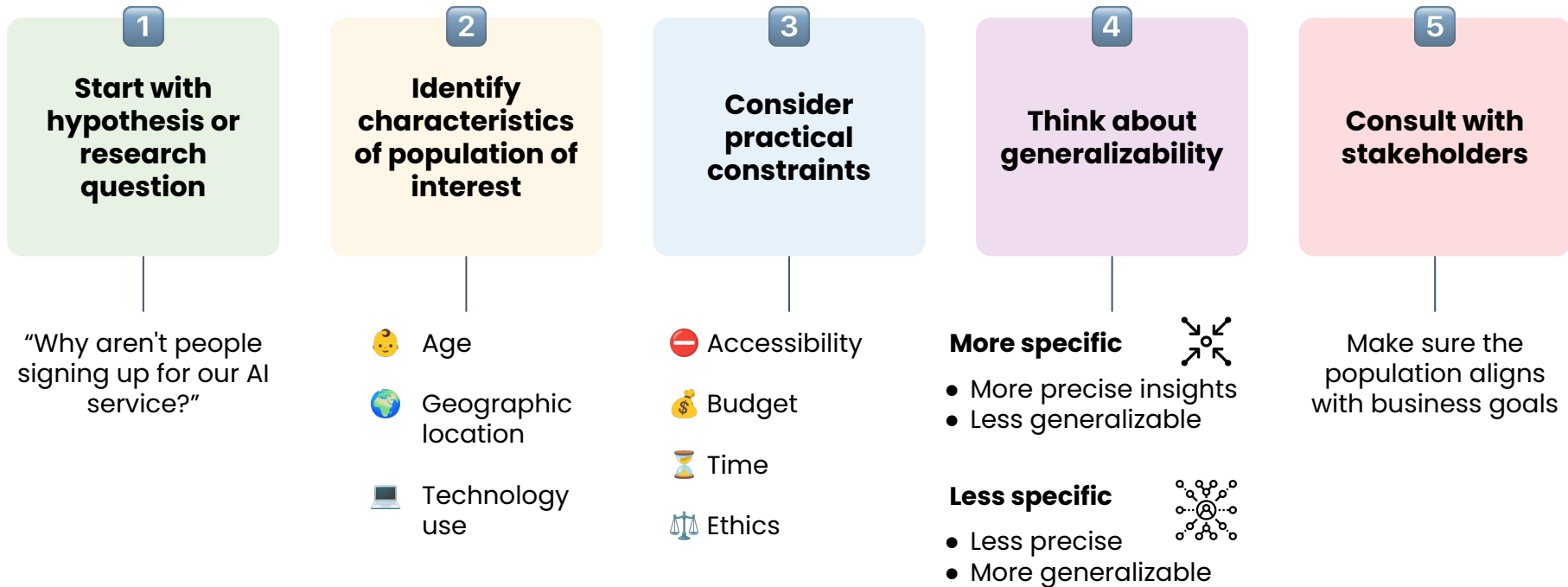


People who use competing services



Recommended process

Population: Internet-using adults aged 18-49 in Canada who currently use at least one AI service, but not ours.



Foundational statistical techniques

Probabilistic samples

Why understand sampling?

- As a data analyst you may:



Work with data that has already been collected



Design samples from scratch

1. Interpret the data correctly

- Identify biases and limitations

2. Communicate results accurately

- Explain how generalizable your conclusions are

3. Collaborate with teams responsible for data collection

- Ensure that best practices are implemented

Scenario

Problem: Understand Canadian public opinion on AI

☁ People who know
AI? 🧑 AI conference!



You
Data Analyst



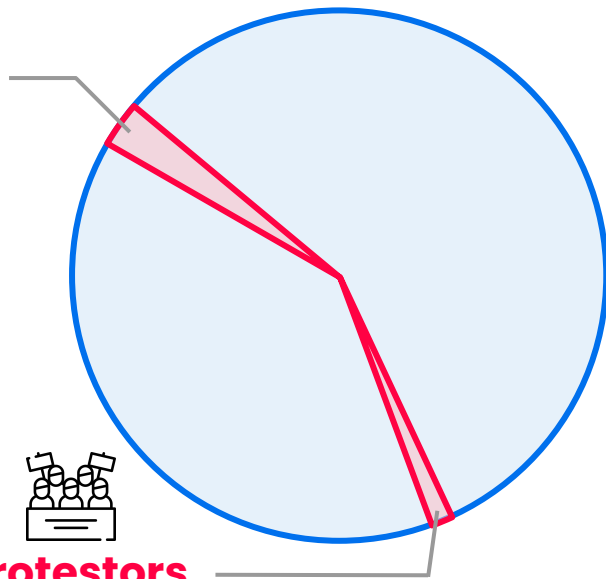
Industry insiders

- Positive views of AI



Both samples, but
not good ones

Population



Protestors

- Skeptical views of AI

Scenario

Goal: To analyze data that represents the entire population you're interested in.

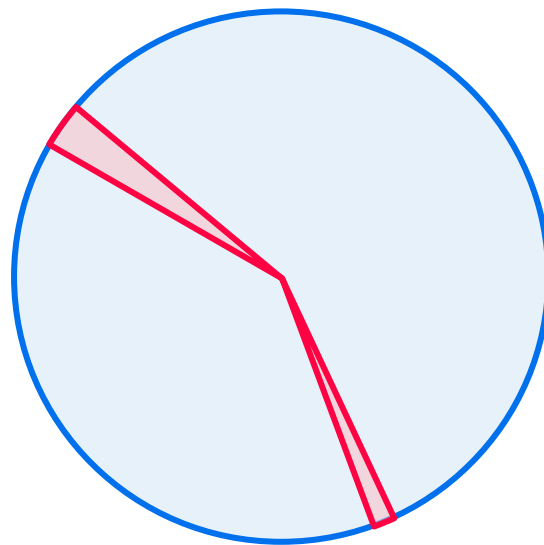
↳ **Adult population of Canada**

Sampling methods:

Probabilistic

Non-probabilistic

Population



Statistical sampling methods

Probabilistic

Give every member of population a **known, non-zero chance** of being selected

★ Potential to be unbiased

1. Simple random
2. Systematic
3. Stratified
4. Cluster

Systematic sampling

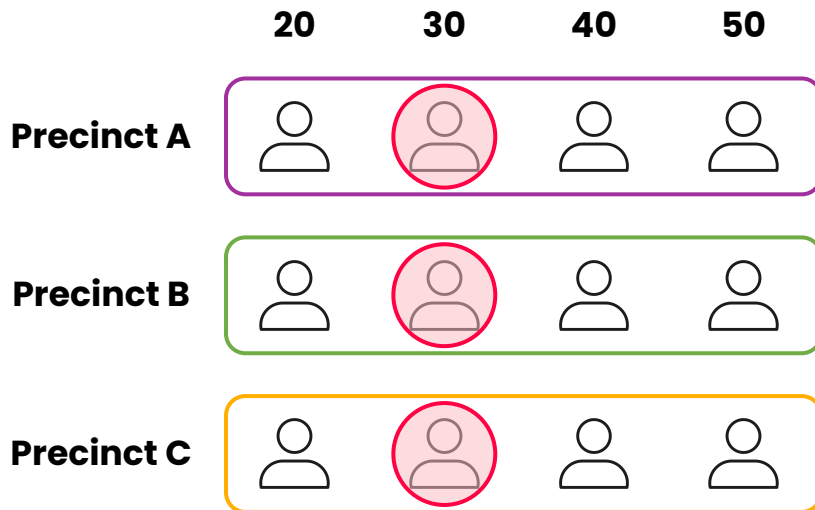
Select **every nth member** of the population after a random starting point

✓ Straightforward

✗ Can cause periodicity bias

If opinions vary by age, this sample will systematically deviate from the population.

Registered voters in Canada



Example:

1. Randomly select one person
2. Choose every 1,000th person after

Simple random sampling

Gives **every member** of the population an **equal** chance of being selected



Straightforward



Provides a good chance of representative sample



Challenging to carry out in large populations

Adult population of Canada

00000	10097	32533	76520	13586	34673	54876	80959	09117	39292	74945
00001	37542	04805	64894	74296	24805	24037	20636	10402	00822	91665
00002	08422	68953	19645	09303	23209	02560	15953	34764	35080	33606
00003	99019	02529	09376	70715	38311	31165	88676	74397	04436	27659
00004	12807	99970	80157	36147	64032	36653	98951	16877	12171	76833
00005	66065	74717	34072	76850	36697	36170	65813	39885	11199	29170
00006	31060	10805	45571	82406	35303	42614	86799	07439	23403	09732
00007	85269	77602	02051	65692	68665	74818	73053	85247	18623	88579
00008	63573	32135	05325	47048	90553	57548	28468	28709	83491	25624
00009	73796	45753	03529	64778	35808	34282	60935	20344	35273	88435
00010	98520	17767	14905	68607	22109	40558	60970	93433	50500	73998
00011	11805	05431	39808	27732	50725	68248	29405	24201	52775	67851
00012	83452	99634	06288	98083	13746	70078	18475	40610	68711	77817
00013	88685	40200	86507	58401	36766	67951	90364	76493	29609	11062
00014	99594	67348	87517	64969	91826	08928	93785	61368	23478	34113
00015	65481	17674	17468	50950	58047	76974	73039	57186	40218	16544
00016	80124	35635	17727	08015	45318	22374	21115	78253	14385	53763
00017	74350	99817	77402	77214	43236	00210	45521	64237	96286	02655
00018	69916	26803	66252	29148	36936	87203	76621	13990	94400	56418
00019	09893	20505	14225	68514	46427	56788	96297	78822	54382	14598
00020	91499	14523	68479	27686	46162	83554				
00021	80336	94598	26940	36858	70297					
00022	44104	81949	85157	47954						
	12550	73742	11100							

Example:

1. **Assign** every adult Canadian a number from 1 to ~30,000,000
2. **Randomly select** 1,000 of those numbers to ask their opinions on AI

Stratified sampling

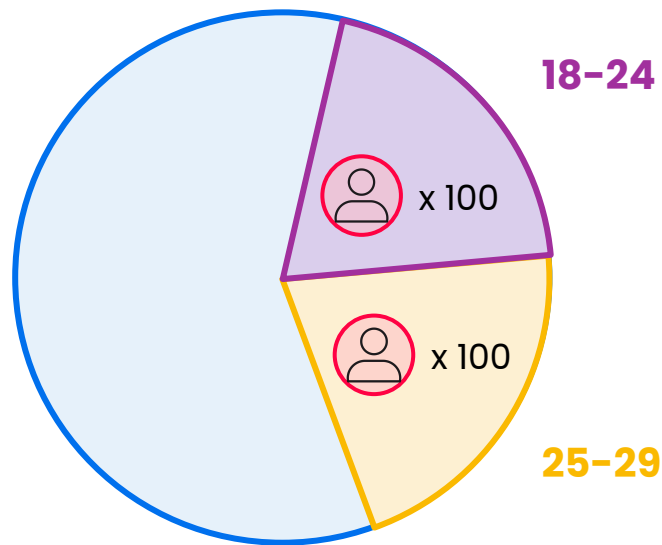
Divides the population into groups (strata), then randomly samples from each group.



Useful when sampling from a diverse set of groups

Common strata:

- Age
- Gender
- Education level
- Product categories
- Price ranges
- Sales regions



Example:

1. Stratify by age group
2. Poll 100 people from each group

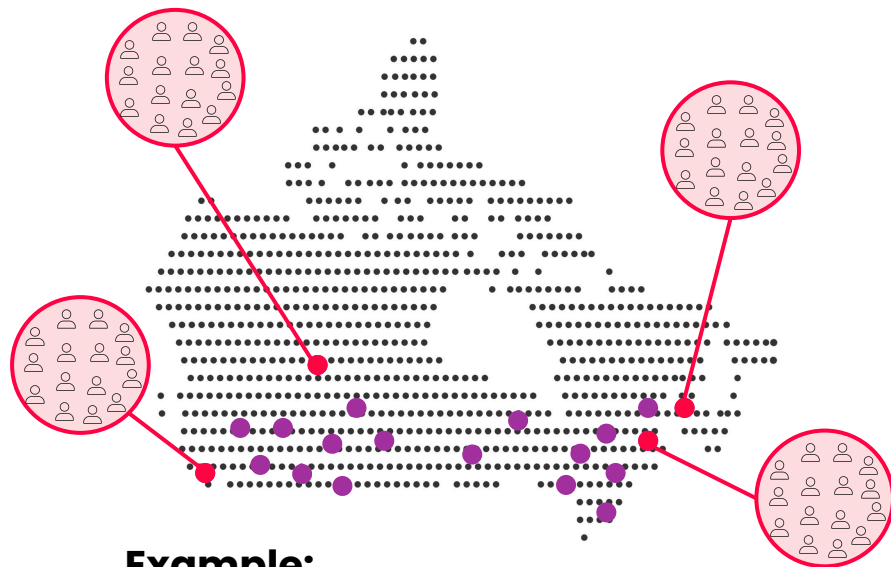
Cluster sampling

Divide the population into clusters, then randomly select entire clusters.

✓ Useful for large, geographically dispersed populations

✓ Efficient

✗ Can introduce cluster-specific bias



Example:

1. Divide Canada into different municipalities
2. Select 10 for your sample, interview randomly selected people



Foundational statistical techniques

Non-probabilistic samples

Statistical sampling methods

Probabilistic

Give every member of population a **known, non-zero chance** of being selected

★ Potential to be unbiased

1. Simple random
2. Systematic
3. Stratified
4. Cluster

Non-probabilistic

Don't give every member of population a known chance of being selected

⚖️ Naturally carry some bias

1. Convenience
2. Quota
3. Judgment
4. Snowball

Convenience sampling

Select members of the population that are easily accessible

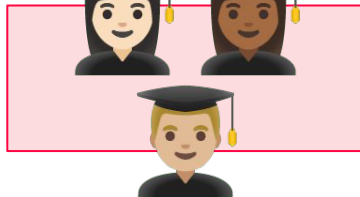
- ✓ Quick
- ✓ Relatively easy

➖ Prone to significant bias

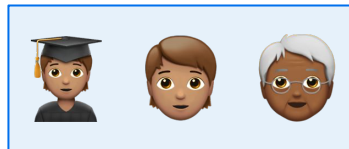
Bias: sample doesn't represent population well

Behavioral research

Easy to recruit



Don't represent population



AI company example



Industry insiders



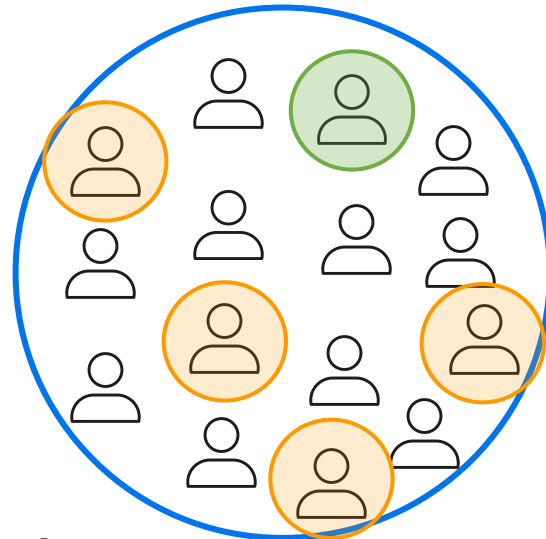
Protestors

Quota sampling

Divide up population, then sample certain number of members from each section

- ✓ Ensures that specific subgroups are represented
- ✓ Faster and easier than stratified sampling

— Introduces bias



Example:

Set quotas for employment in Canada

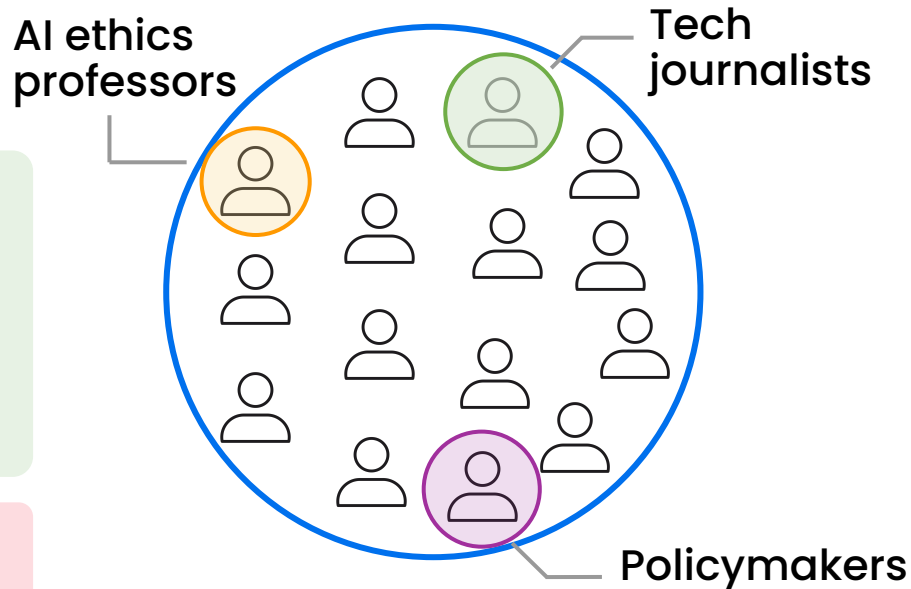
- **87 people** who are working
- **13 people** who are not working

Judgment sampling

Relies on researcher's expertise

- ✓ Useful when you need insights from specific types of individuals
- ✓ Target individuals with highly relevant knowledge

— Highly subjective



Snowball sampling

Start with small group of subjects, each of whom then recruits future subjects



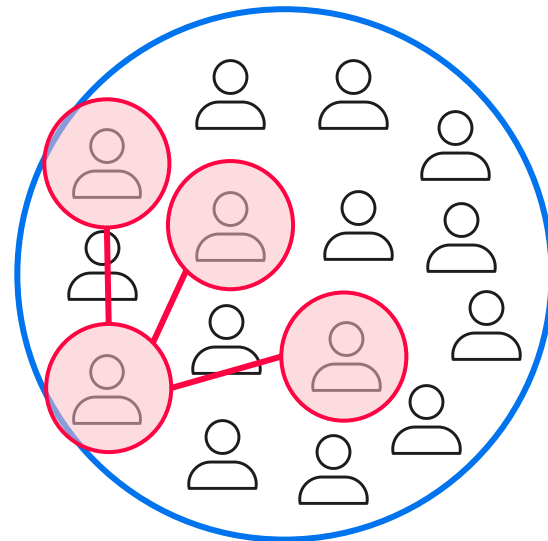
Useful for studying hard-to-reach populations



Limited by social networks of initial subjects



Potentially missing important segments of the population



Example:

1. Start with people who've opted out of AI tools
2. Ask them to refer others

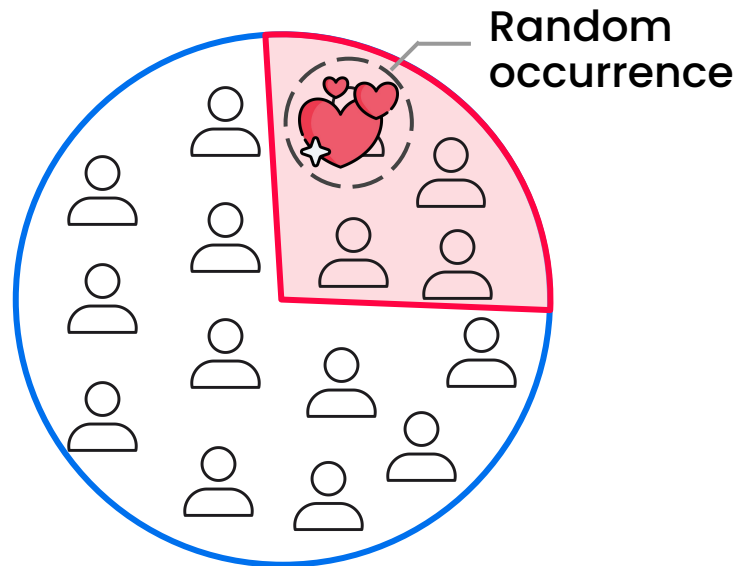
Foundational statistical techniques

Types of bias

What is bias?

A **systematic** difference between sampling method and population

- Problem occurs in a predictable, likely preventable way



What is bias?

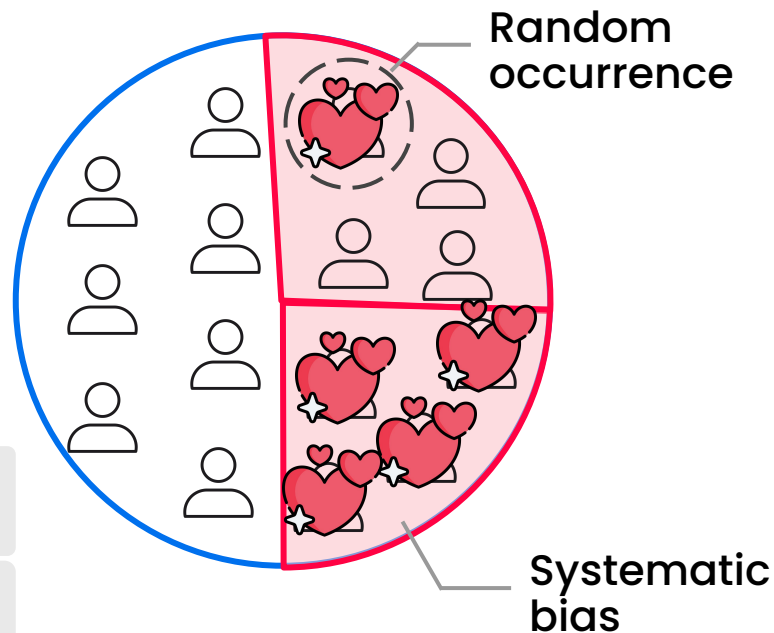
A **systematic** difference between sampling method and population

- Problem occurs in a predictable, likely preventable way
- Often the result of deviating from probabilistic sampling
- Types of bias:

• Sampling bias	When deciding how to sample
-----------------	------------------------------------

• Measurement bias	When data is collected
• Response bias	

• Analysis bias	During insight-finding
-----------------	-------------------------------

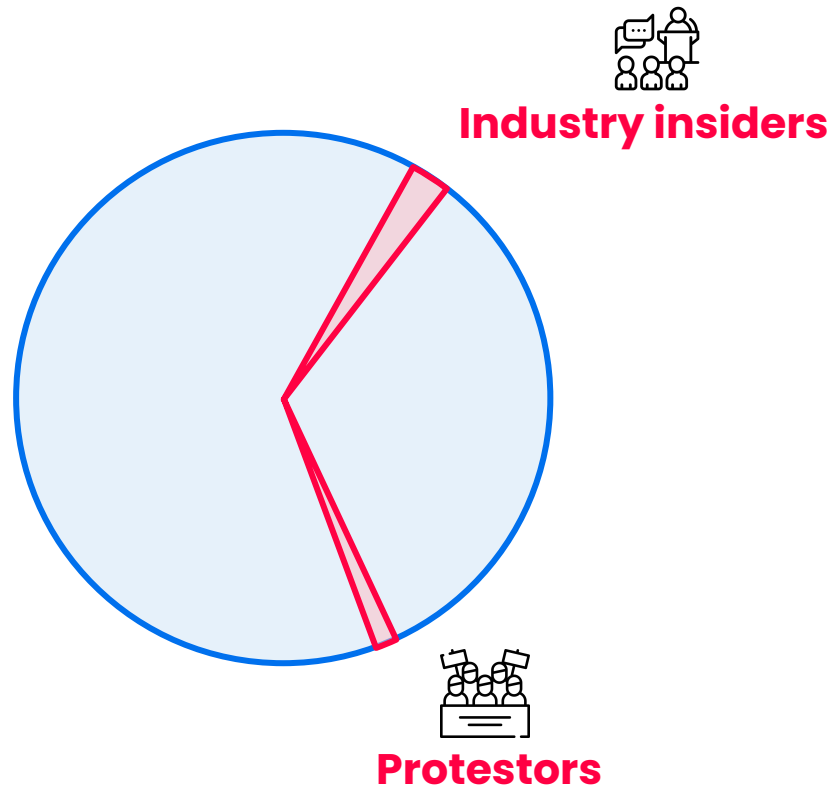


Sampling bias

Sample doesn't accurately represent the population of interest



Unlikely those groups represent views of most Canadians

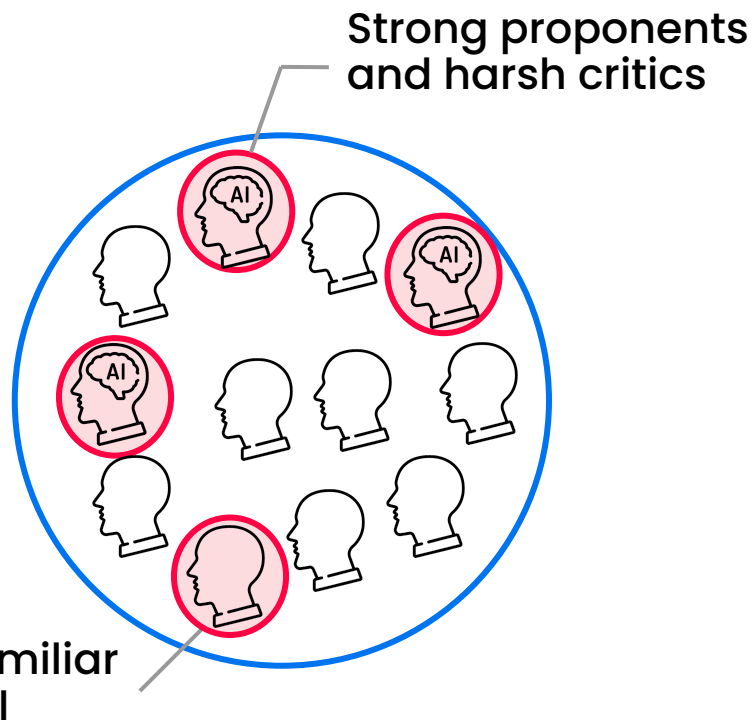


Selection bias

Sample is selected in a nonrandom way

- ❌ Mismatch between sample and population of interest
- ❌ Some groups are under or overrepresented

- ✅ Use probabilistic sampling methods
- ✅ Avoid over- or under-sampling any particular group
- ✅ Be transparent about limitations



Nonresponse bias

- Common in samples of people

- ✓ Consider sending follow-ups
- ✓ Include small incentives for participation



Smartphone apps may ask a user to rate their app



Users can dismiss the notification



People that do leave a review:

- Aren't a random sample
- Often have positive view of the app

Instrument bias

Comes from having faulty equipment or a poorly designed survey

- ✓ Use good quality tools
- ✓ Use well-worded surveys
- ✓ Take multiple measurements

Example: In 2020, Fitbit...



Replaced smartwatches free due to issue with heart sensor

Example: As simple as asking...



"You agree that pineapple belongs on pizza, right?"

Observer bias

Occurs when the person measuring lets expectations affect what they see



Might unconsciously focus on positive feedback

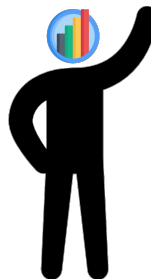


Have multiple people take the measurement

For example:



Action: Rolling out dark mode



Expectation: Feature will be well-reviewed

Response bias

- Observed in samples of people
- People may not be completely forthcoming



Questions like:

- ❌ “How much money do you make?”
- ❌ “When was the last time you dipped your french fries in mayonnaise?”

You can try:

- ✅ Emphasizing how important honest answers are
- ✅ Designing your questions to be as objective as possible

Analysis bias

Way you perform your analysis can lead to results that don't reflect truth

— **Confirmation bias:** looking for evidence for something you already believe

- ✓ Stay open-minded
- ✓ Let the data tell its own story



- **Hope:** A new feature is improving in user retention
- **Analysis bias:** Focus on the 2 metrics that show most positive results

Foundational statistical techniques

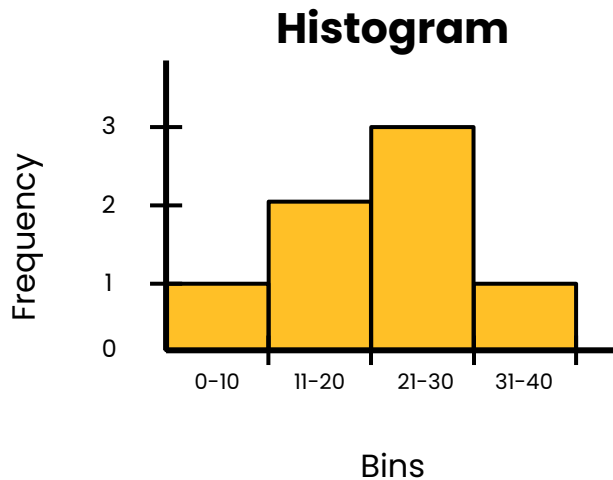
Histograms

Visualizing distributions

- A **distribution** tells you how often different values occur

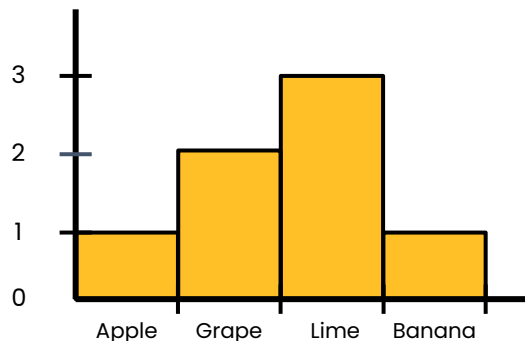
A histogram:

- Aggregates numerical data into bins
- Visualizes frequency of values within bins
- Bins make it easier to see overall distribution



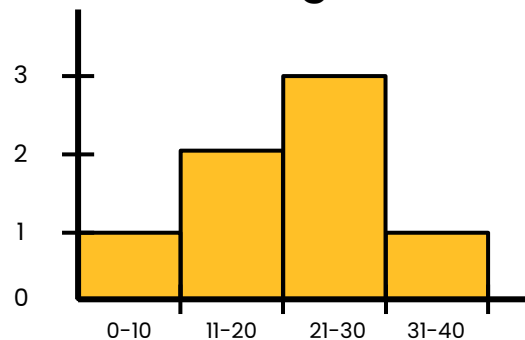
Visualizing distributions

Column Chart



- **x-axis:** categorical or discrete numerical feature
- **y-axis:** numerical feature

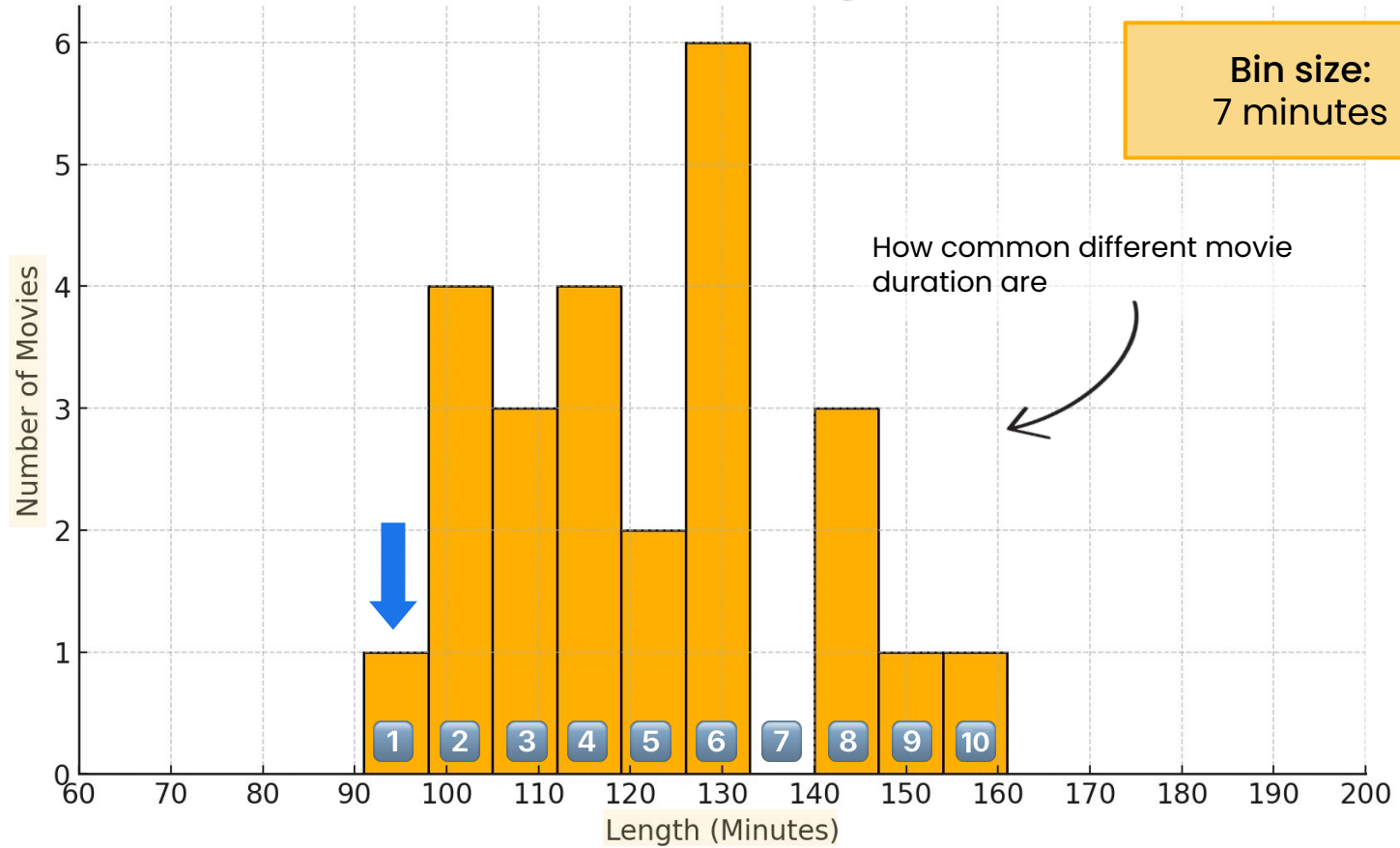
Histogram



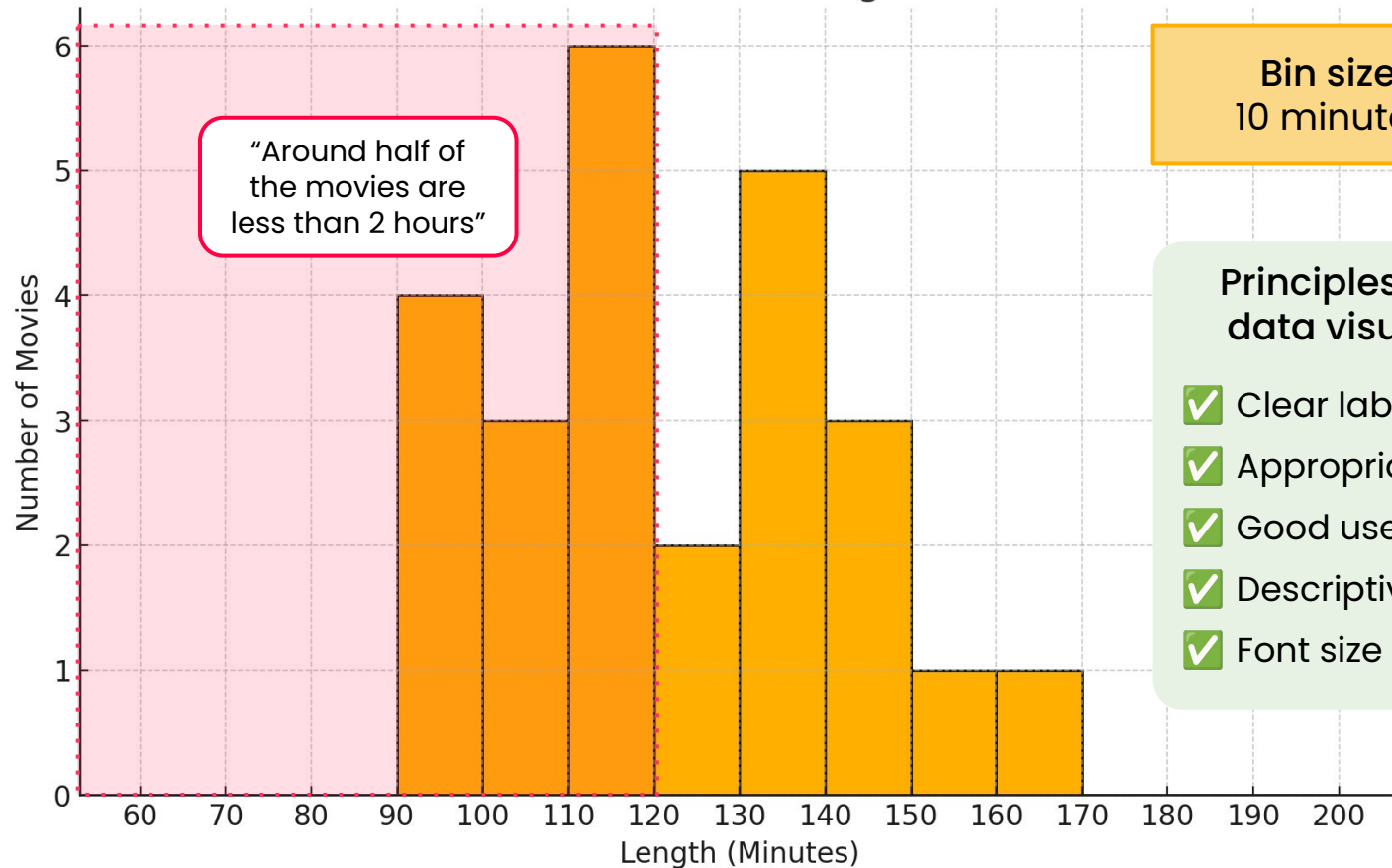
- **x-axis:** bins that group **continuous** numerical values
- **y-axis:** frequency

Distribution of Movie Lengths in 2013

Bin size:
7 minutes



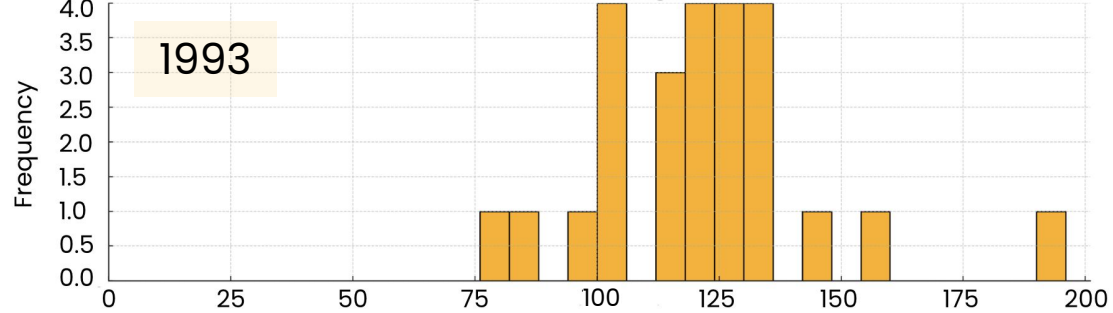
Distribution of Movie Lengths in 2013

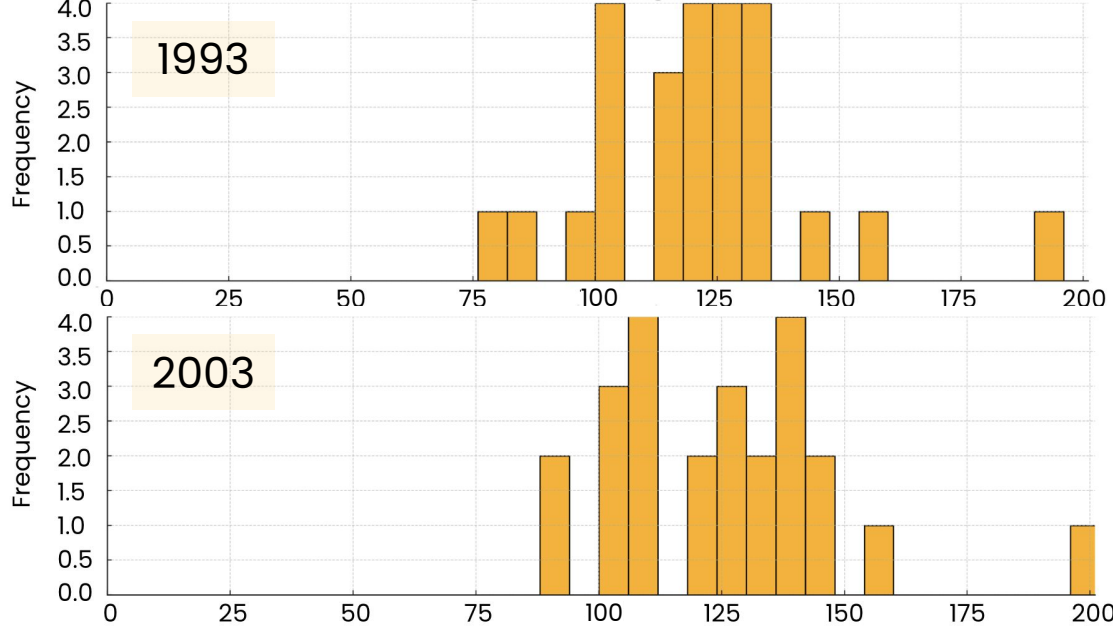


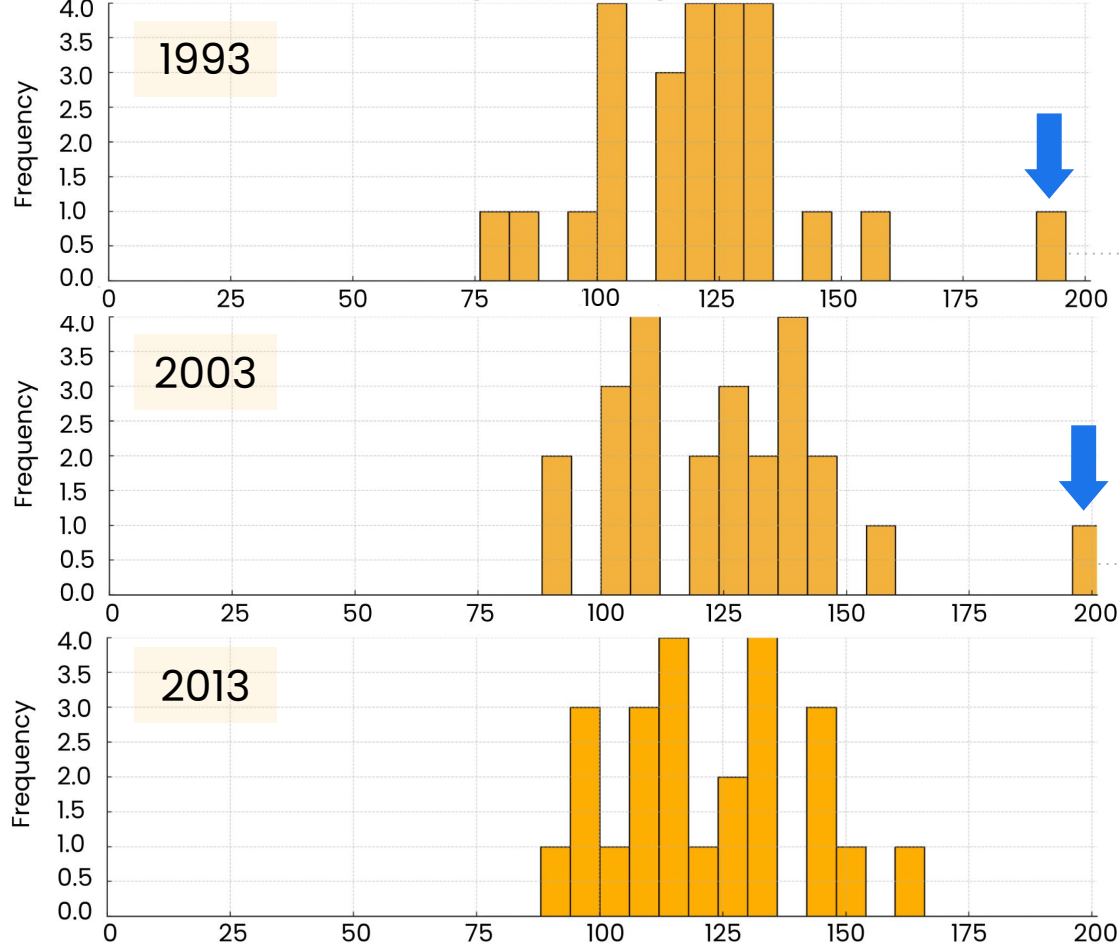
Bin size:
10 minutes

Principles for good data visualization

- ✓ Clear labels
- ✓ Appropriate scales
- ✓ Good use of color
- ✓ Descriptive title
- ✓ Font size is legible







Schindler's List



**Lord of the Rings:
Return of the King**



Foundational statistical techniques

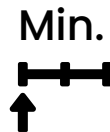
Demo: Plotting distributions



Foundational statistical techniques

Central tendency,
variability, and skewness

Characterizing a distribution



Central tendency

- Where is the middle of the data?
- What values are most frequent?
- Common statistics include:
 - Mean
 - Median
 - Mode

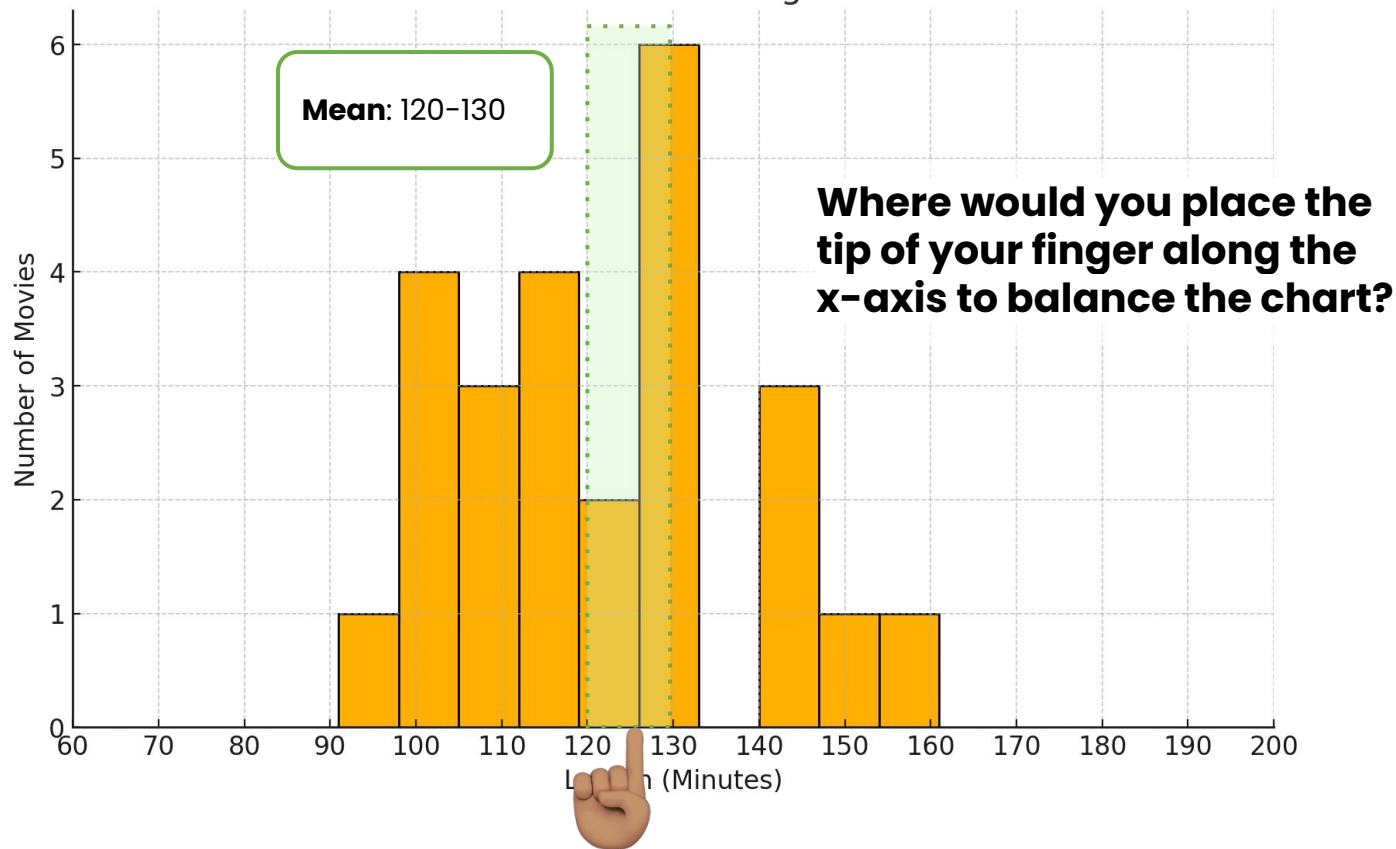
Variability

- How different are the values in the data?
- Are they clustered together or spread out?
- Measures of variability:
 - Range
 - Interquartile range
 - Variance
 - Standard deviation

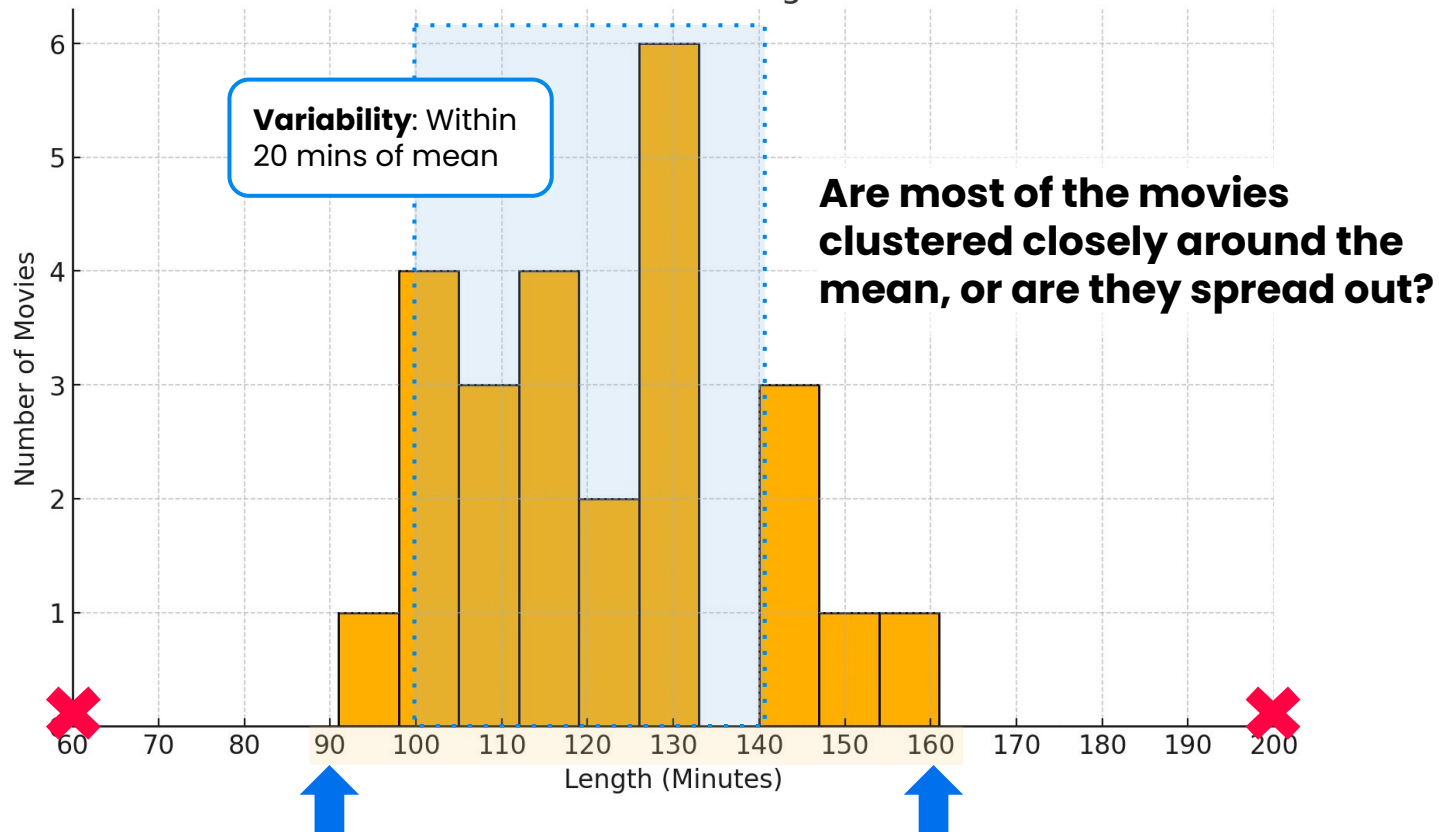
Skewness

- Is the distribution symmetrical or does it lean to one side?
- Compare:
 - Mean
 - Median
- Calculate it directly in a spreadsheet

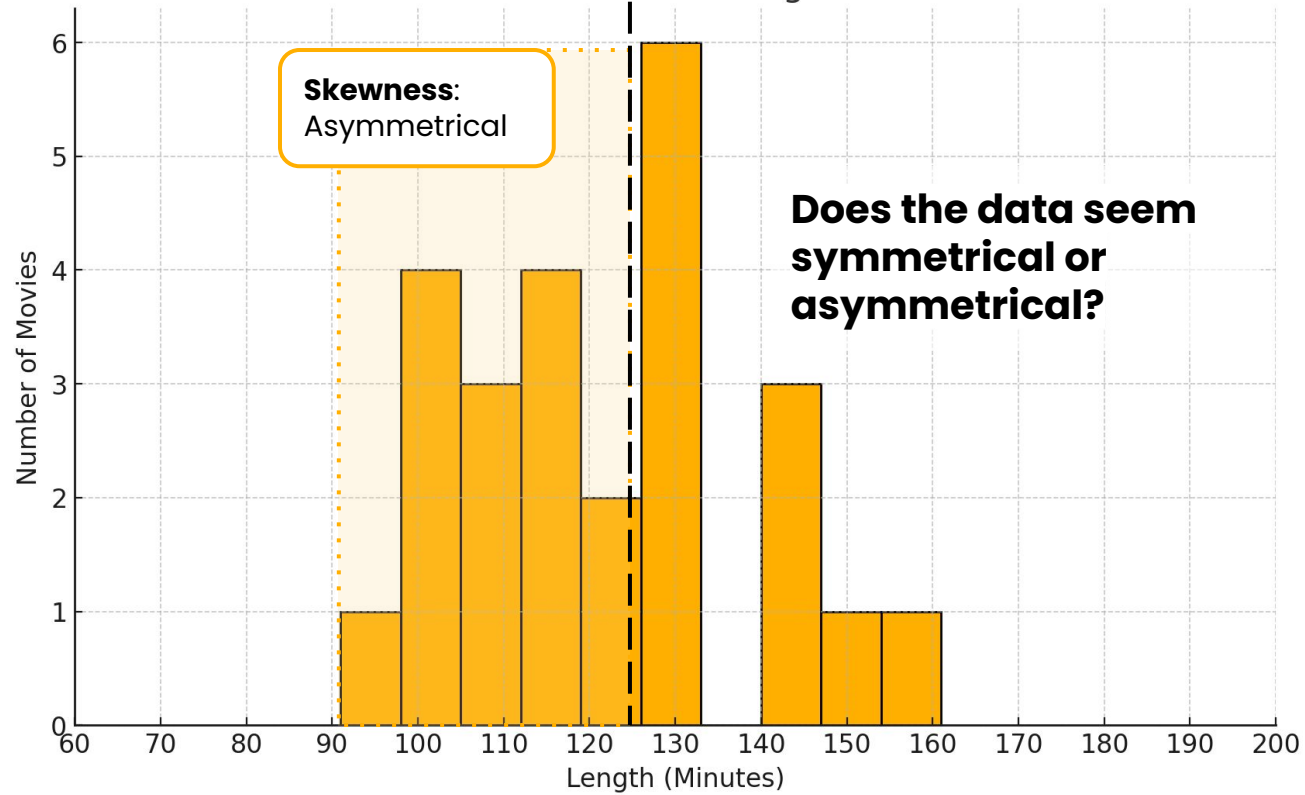
Distribution of Movie Lengths in 2013



Distribution of Movie Lengths in 2013



Distribution of Movie Lengths in 2013





Foundational statistical techniques

Central tendency:
mean and mode

Mathematical notation

X

Specific value
in your sample

Film	Duration
Man of Steel	143



$x = 143$

n

Size of the sample



$n = 10$



$n = \# \text{ of rows}$

Mean

x = specific movie durations

$n = 25$ 

To calculate the **mean movie duration**:

	Film	Length
x_1 →	Man of Steel	143
	...	...
x_{25} →	A Good Day to Die Hard	98

$$\bar{x} = \frac{\text{Sum of all values}}{\text{Number of values}} = \text{Average of the values} = \frac{3035 \text{ minutes}}{25 \text{ movies}} = \sim \mathbf{121 \text{ minutes}}$$

Σ means add all x_i values up

i is in an index that ranges from 1 until n

$$= \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \dots + x_{25}}{n} = \frac{3035 \text{ minutes}}{25 \text{ movies}} = \mathbf{121.4 \text{ minutes}}$$

Mode

Most common value

Movie lengths in 2013:

91 98 98 98 100 107 109 110 112 113 115 116 119 124 126 130 130 130 131 132 143 143 146 153 161

Mode

Used for **discrete numerical** data or **categorical** data

- Mean and median aren't applicable

Candidate	% of Votes
A	55%
B	30%
C	15%



No way to calculate the **mean vote**



Center of mass: Candidate A, who won the **most votes**



Mode is useful!



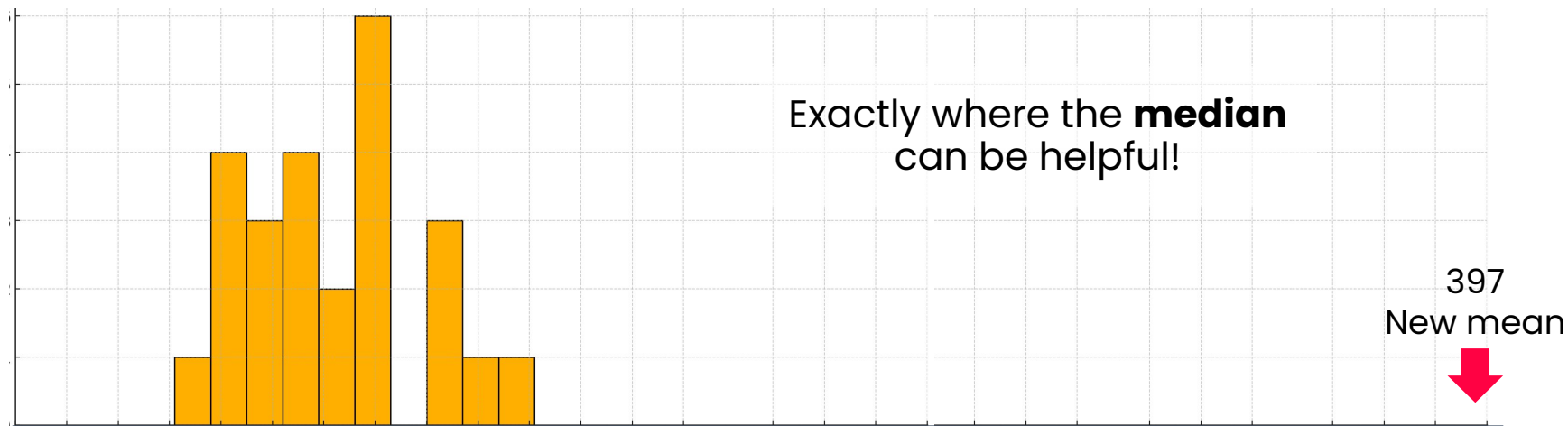
Foundational statistical techniques

Central tendency: median

Mean

- Useful when distribution is **not** strongly skewed
- **Sensitive** to large and small values

$$\frac{143 + 130 + 116 + \dots + 7000}{25 \text{ movies}} = 397 \text{ minutes}$$



Median

The value in the center of the dataset

1. Line up all the values x_i in ascending order
2. Cross them out in pairs



Median

- Helpful when your sample is skewed or contains **outliers**
- Outliers: extreme values in your data

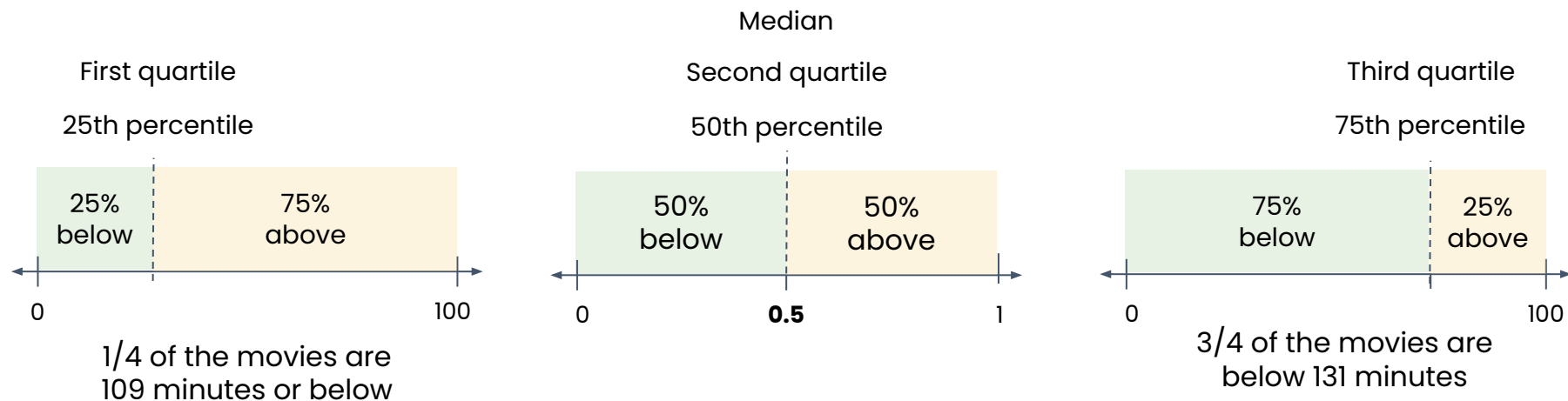
$n = 26$



$$\frac{119 + 124}{2} = 121.5 \text{ minutes}$$

Percentiles

- Useful when dealing with highly unusual values or skewed data





Foundational statistical techniques

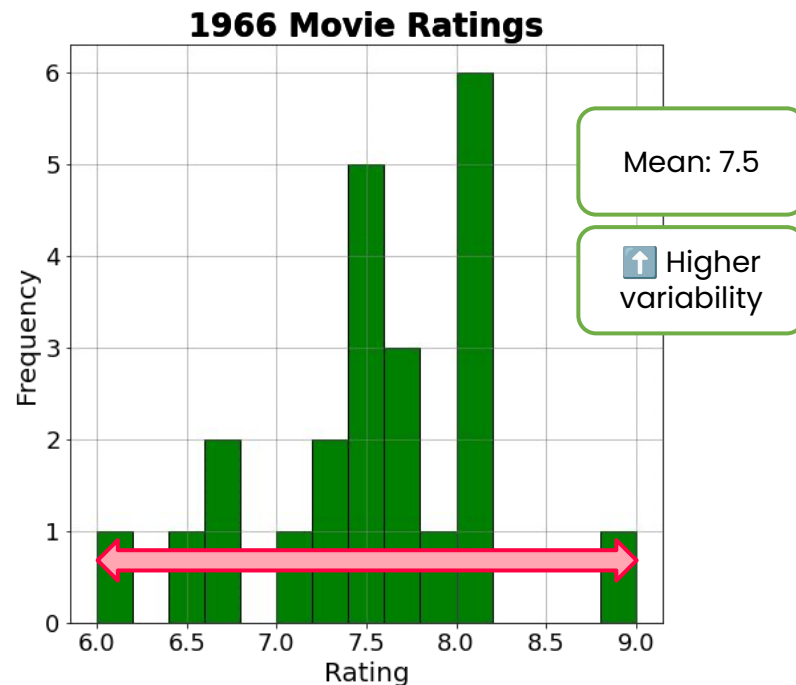
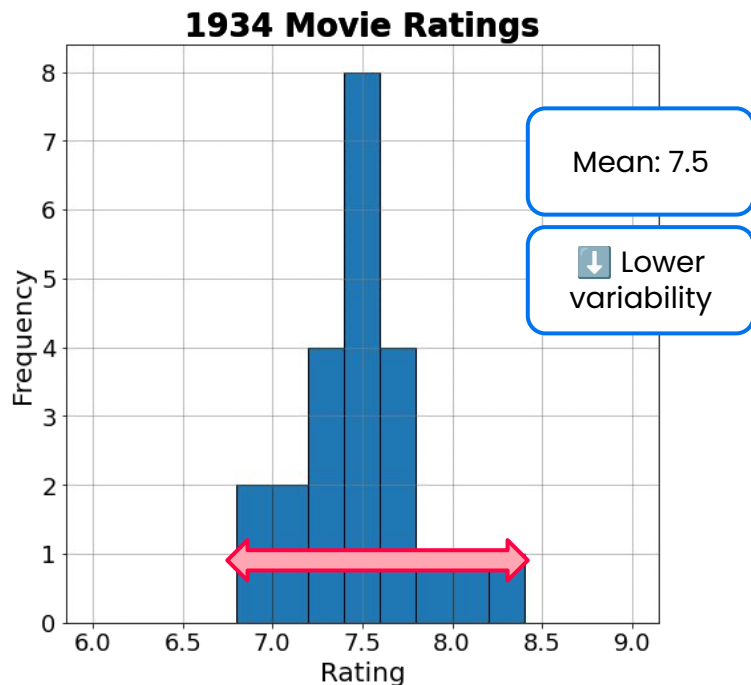
Demo: central tendency

Foundational statistical techniques

Variability: range and
interquartile range

Intuition

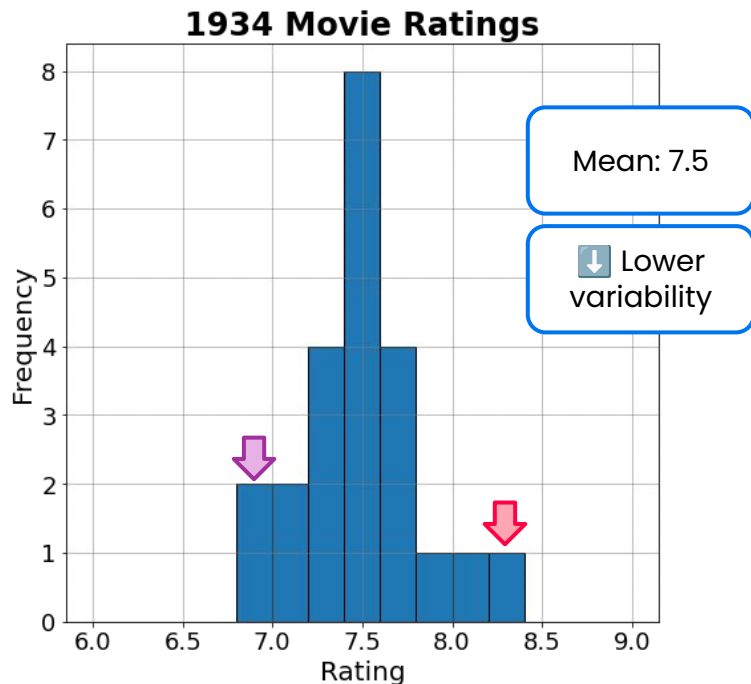
Ratings on IMDb: 1 to 10 scale



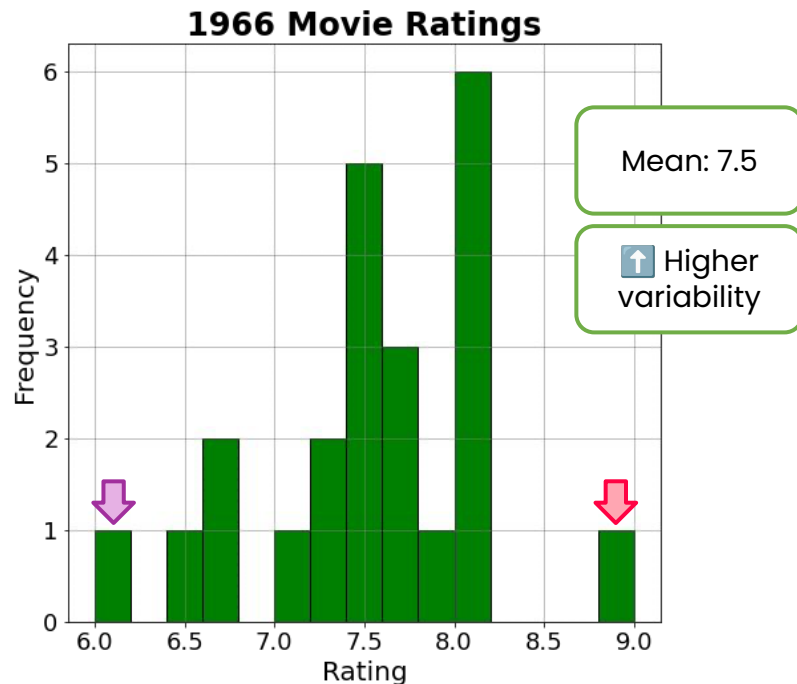
Range

- Largest distance between any two values

$$\text{range} = \text{max} - \text{min}$$



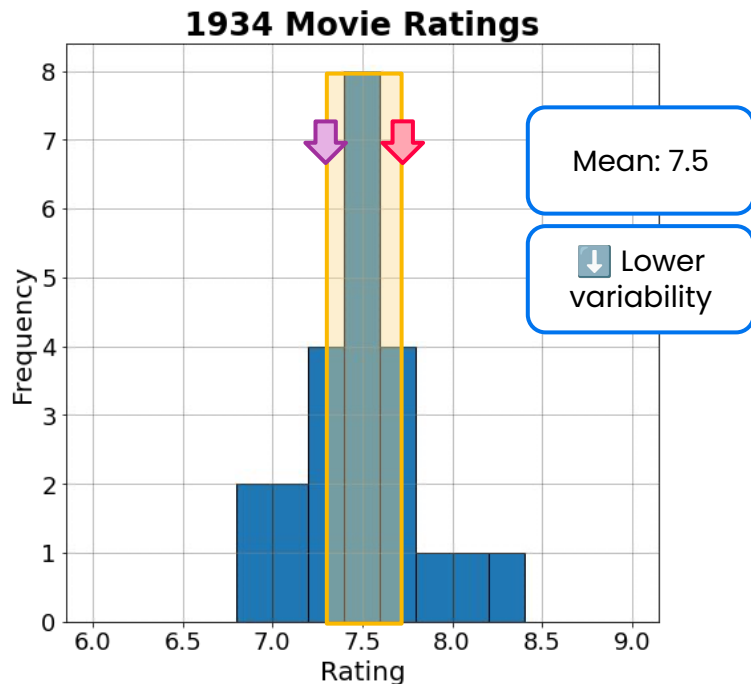
$$8.3 - 6.9 = 1.4$$



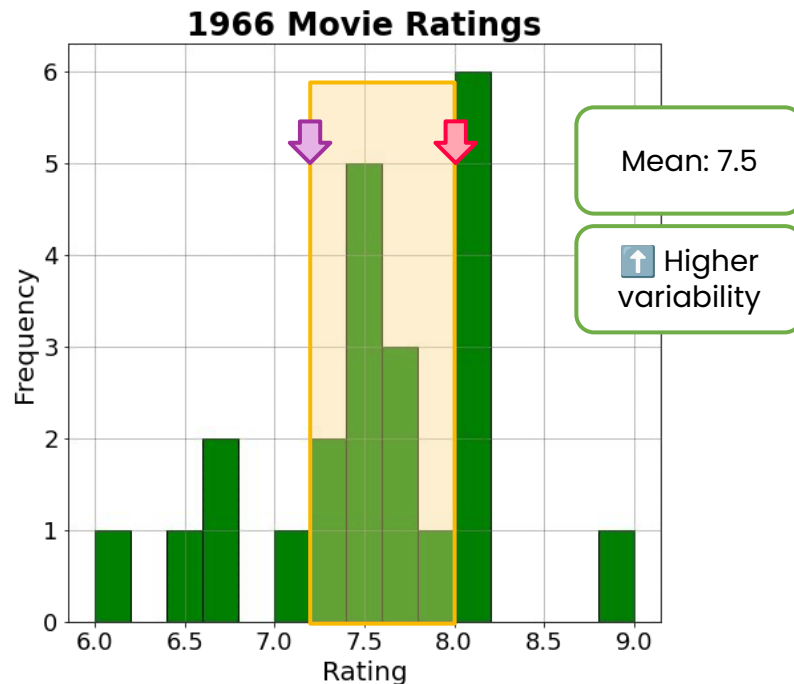
$$9.0 - 6.1 = 2.9$$

Interquartile Range (IQR)

$$\text{IQR} = Q3 - Q1$$



$$7.75 - 7.25 = 0.5$$



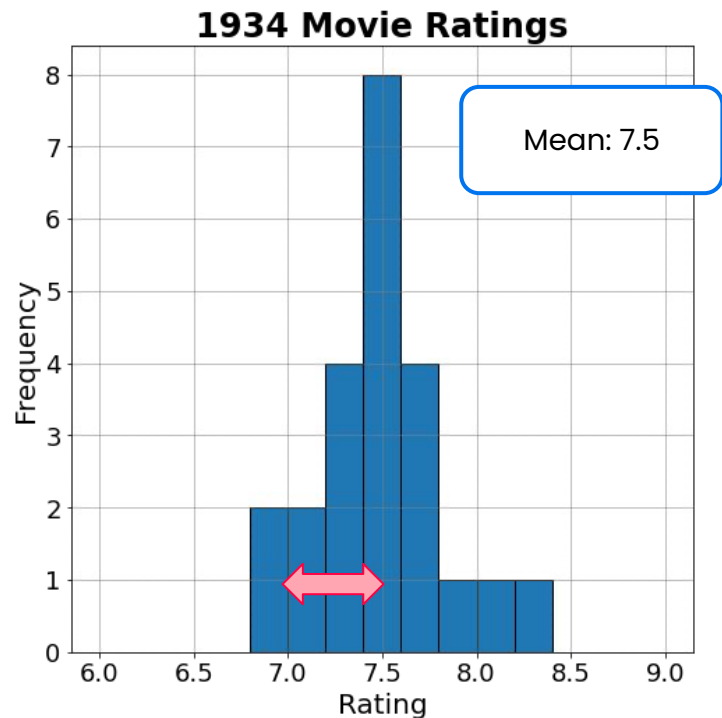
$$8 - 7.25 = 0.75$$



Foundational statistical techniques

Variability: variance and
standard deviation

Sample variance

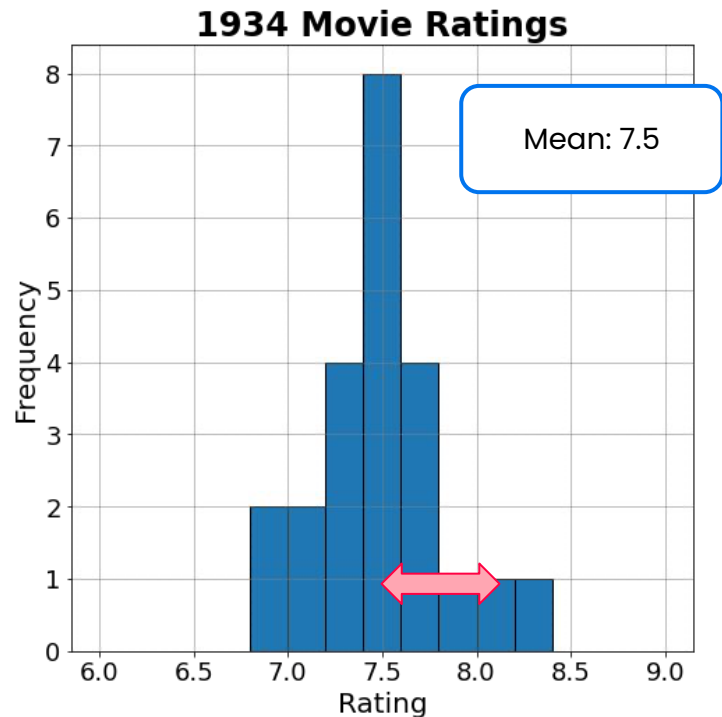


Idea behind variance: find the average squared distance of values from the mean

Movie title	Rating
Cleopatra	6.9
It Happened One Night	8.3
...	...

$$(\longleftrightarrow)^2$$

Sample variance



Idea behind variance: find the average distance of values from the mean

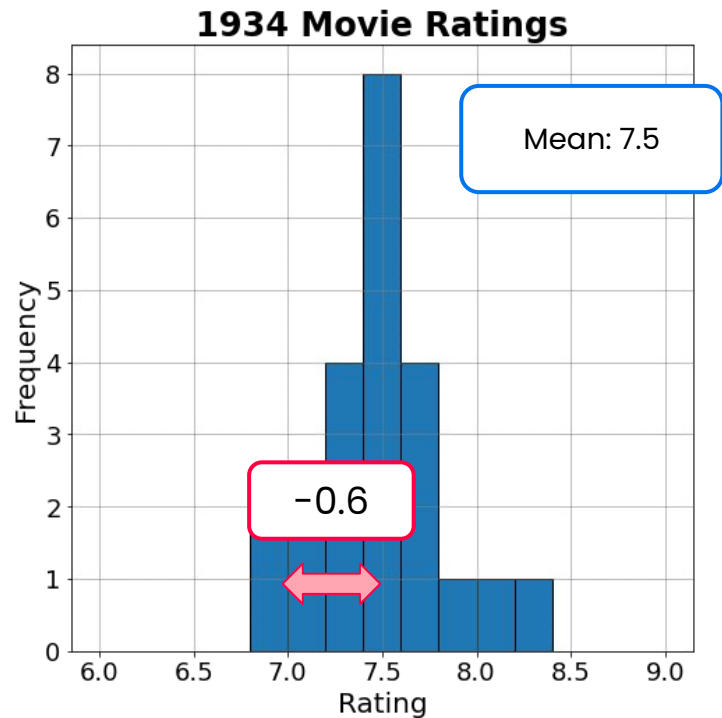
Movie title	Rating
Cleopatra	6.9
It Happened One Night	8.3
...	...

Squared
deviations

$$(\text{distance})^2 + (\text{distance})^2 + (\dots)^2$$

$$= \frac{\text{sum of squared deviations}}{n - 1} = \text{Sample variance}$$

Sample variance

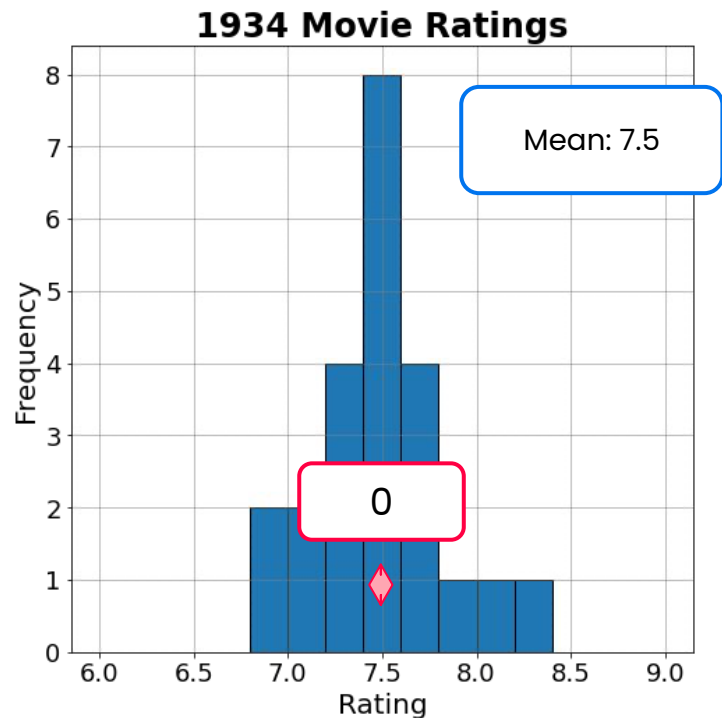


Idea behind variance: find the average distance of values from the mean

Movie title	Rating
Cleopatra	6.9
The Merry Widow	7.5
It Happened One Night	8.3

0.36

Sample variance

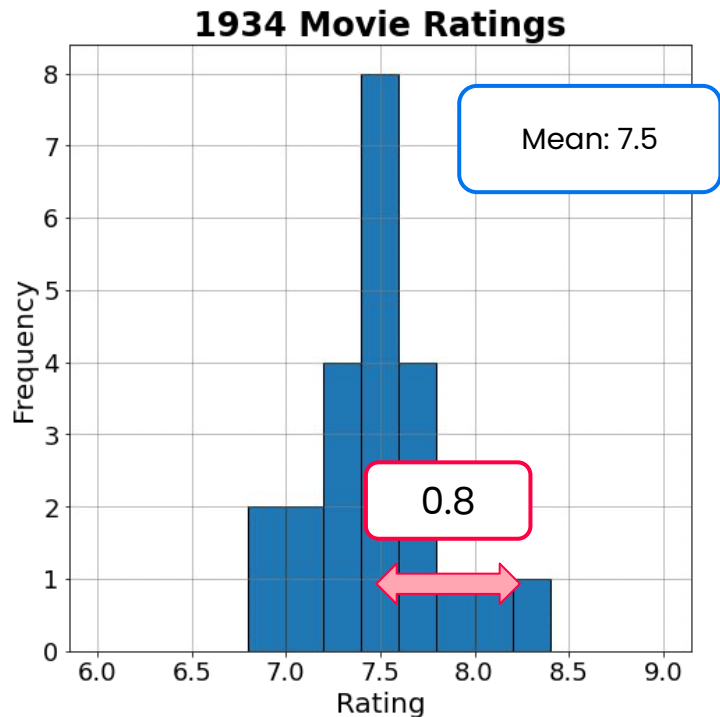


Idea behind variance: find the average distance of values from the mean

Movie title	Rating
Cleopatra	6.9
The Merry Widow	7.5
It Happened One Night	8.3

$$0.36 + 0.0$$

Sample variance



Idea behind variance: find the average distance of values from the mean

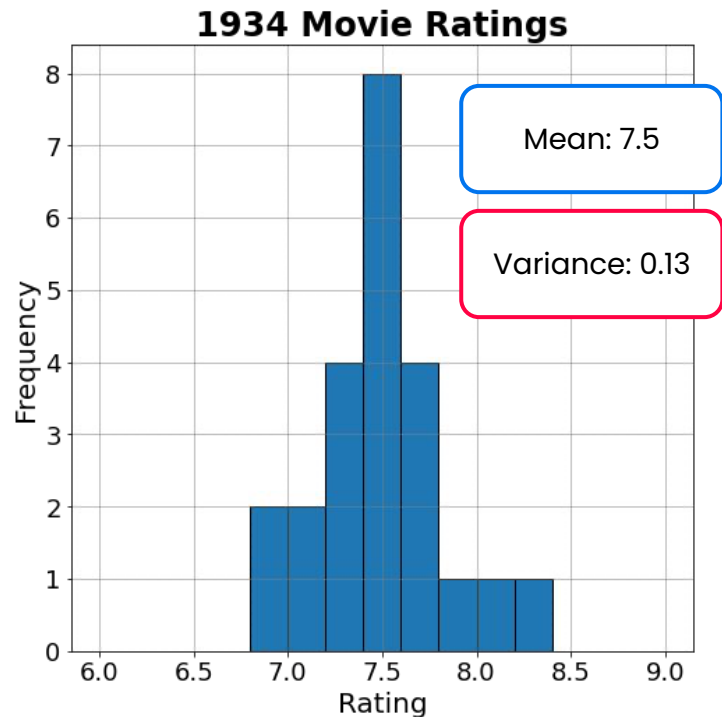
Movie title	Rating
Cleopatra	6.9
The Merry Widow	7.5
It Happened One Night	8.3

0.36 + 0.0 + 0.64 + ...

Purposes of squaring

- Positives and negatives don't cancel out
- Emphasizes larger deviations

Sample variance



Idea behind variance: find the average distance of values from the mean

Movie title	Rating
Cleopatra	6.9
The Merry Widow	7.5
It Happened One Night	8.3

0.36 + 0.0 + 0.64 + ...

$$\frac{2.91}{n - 1} = \frac{2.91}{23} = 0.13$$

Why $n - 1$?

- Minor adjustment to make calculation accurate for a sample
- Makes estimate more reliable for smaller sample sizes

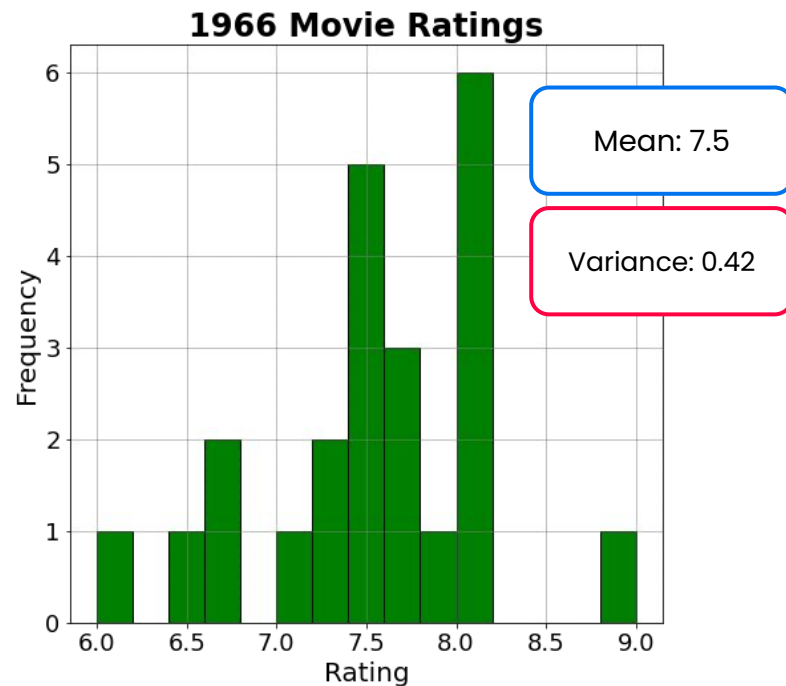
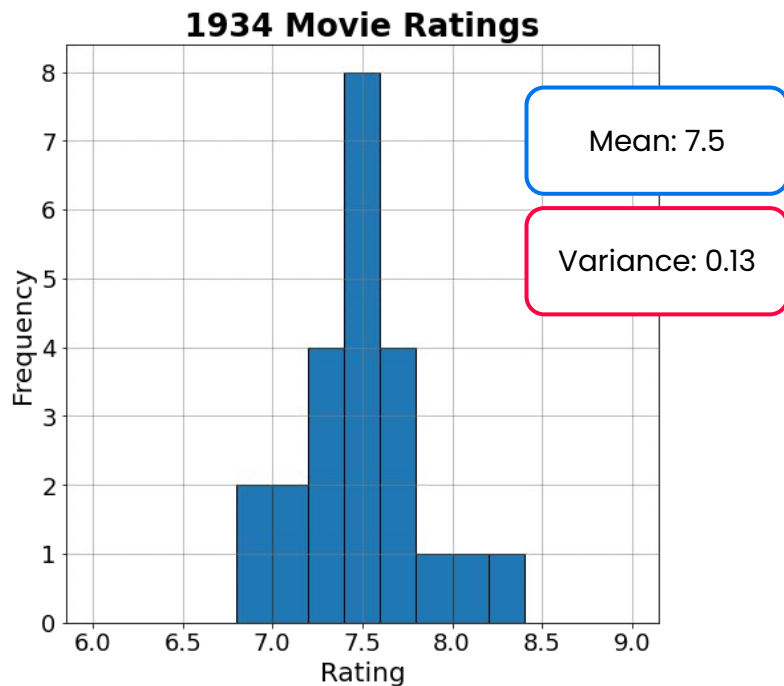
Idea behind variance: find the average distance of values from the mean

Movie title	Rating
Cleopatra	6.9
The Merry Widow	7.5
It Happened One Night	8.3

0.36 + 0.0 + 0.64 + ...

$$\frac{2.91}{n - 1} = \frac{2.91}{23} = 0.13$$

Sample variance



Formula for sample variance

$$\frac{\sum (x_i - \bar{x})^2}{n - 1}$$

Not in the same units
as your data

Formula for standard deviation

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

= 0.355 points

Sample **mean**

$\bar{x}$

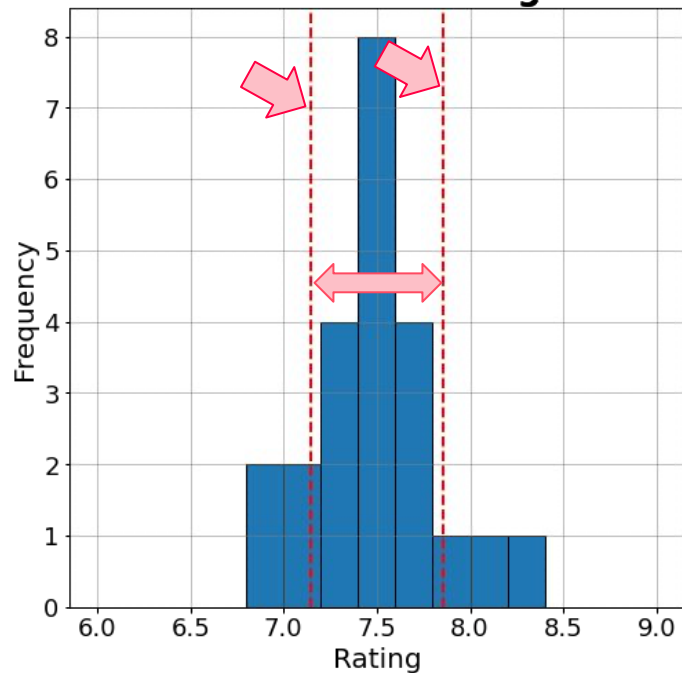
Sample **standard deviation**

s

Sample **variance**

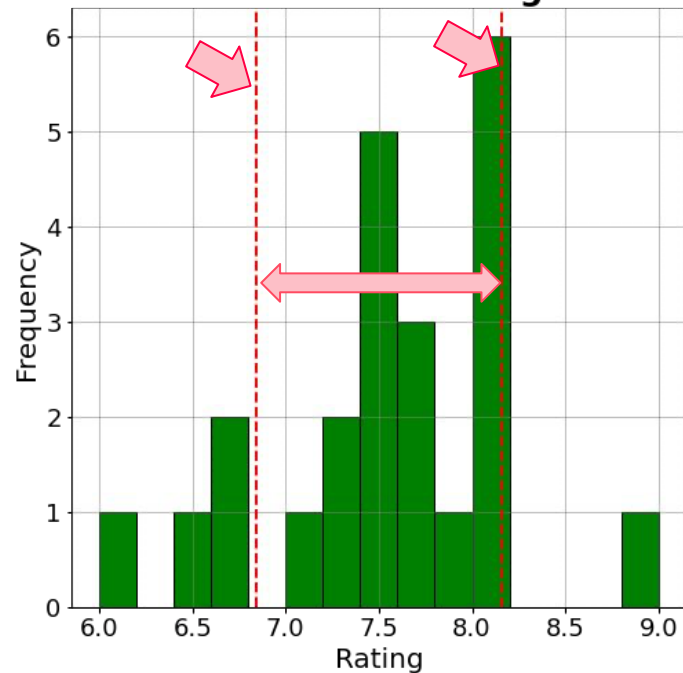
s^2

1934 Movie Ratings



$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} = 0.355$$

1966 Movie Ratings



$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} = 0.655$$

Foundational statistical techniques

Skewness

Skewness

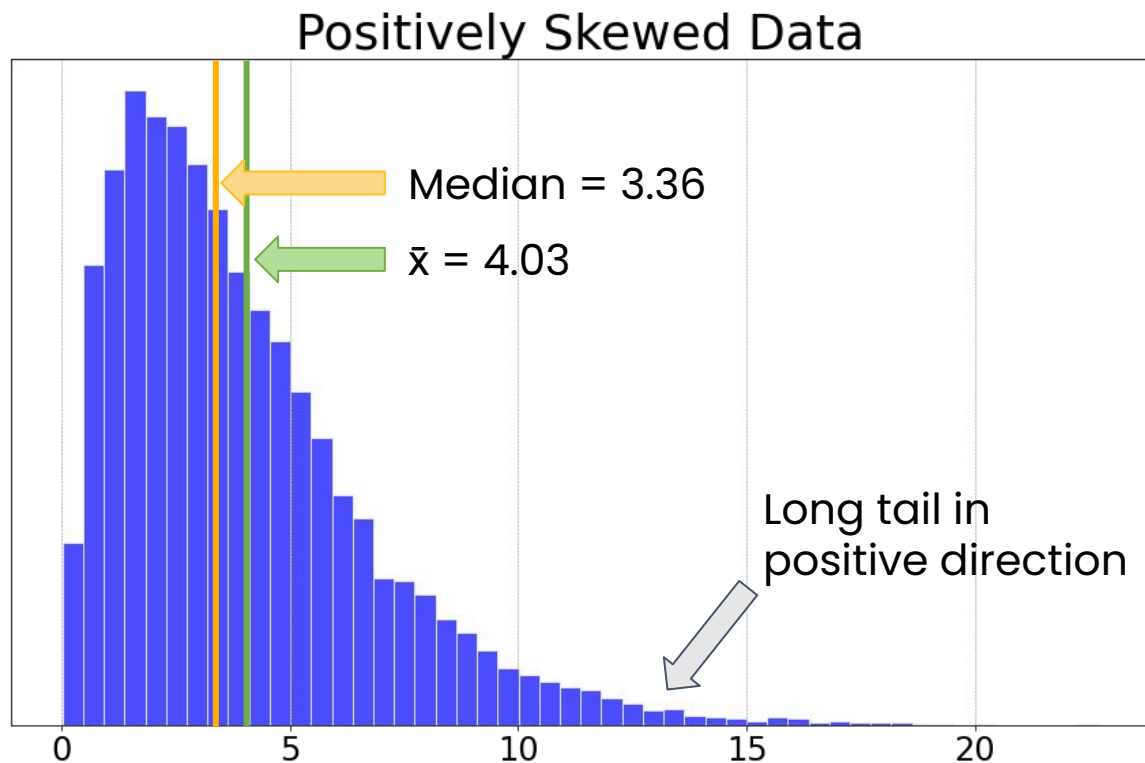
- **Mean:** very sensitive to extreme values
- **Median:** relatively insensitive to unusual values
- **One way to estimate skew:** compare the mean and median
- When values are **asymmetrical: mean** will move further towards the tail

$$\frac{143 + 130 + 116 + \dots + 7000}{25 \text{ movies}} = 397 \text{ minutes}$$

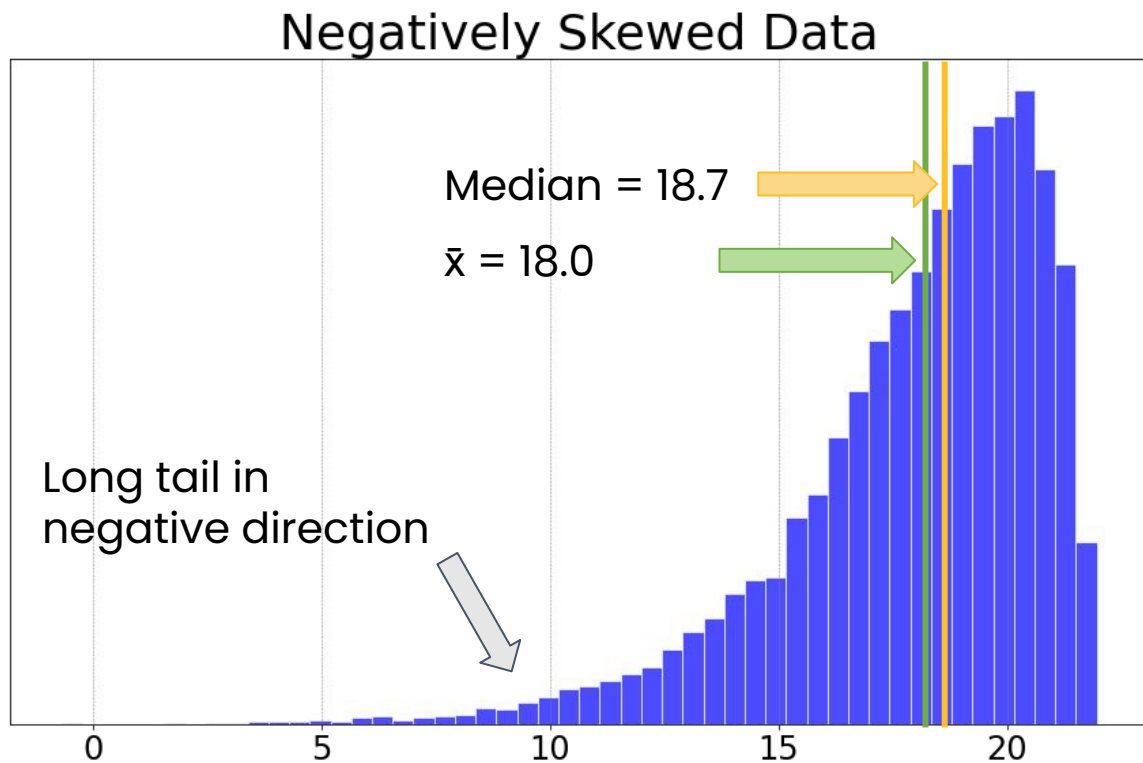

91 98 98 ... 119 124 ... 153 161 7000

The diagram illustrates the impact of an outlier (7000) on the mean. The mean calculation is shown as (143 + 130 + 116 + ... + 7000) / 25 movies = 397 minutes. Below this, a sequence of values is shown: 91, 98, 98, ..., 119, 124, ..., 153, 161, 7000. The values 91, 98, 98, 153, 161, and 7000 are crossed out with red slashes. The values 119 and 124 are circled in green, and a green arrow points from the 7000 in the mean calculation down to the 124 in the data sequence, highlighting how the outlier affects the mean.

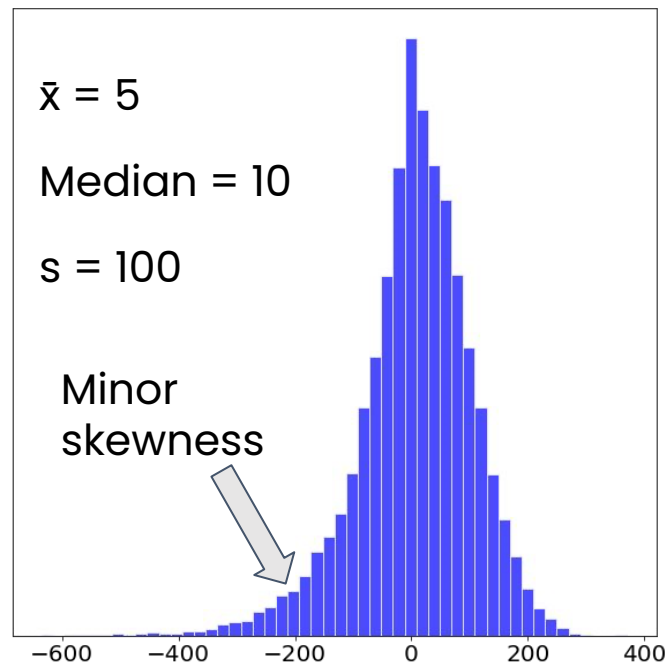
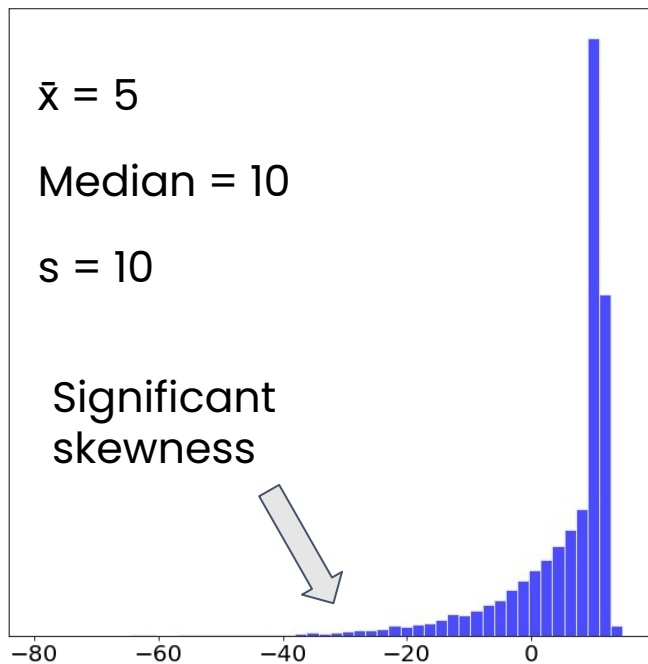
Skewness



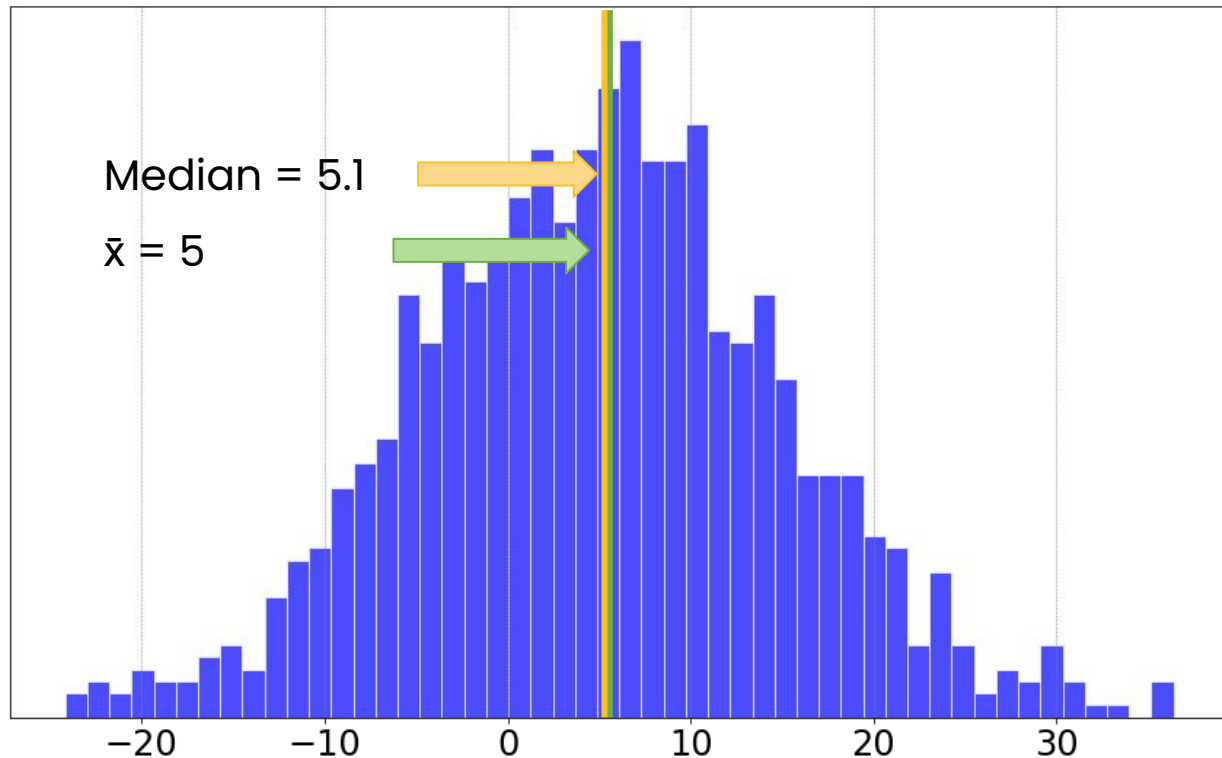
Skewness



Skewness



Skewness



Quick test!

- $\bar{x} = 18$
- Median = 10
- $s = 15$

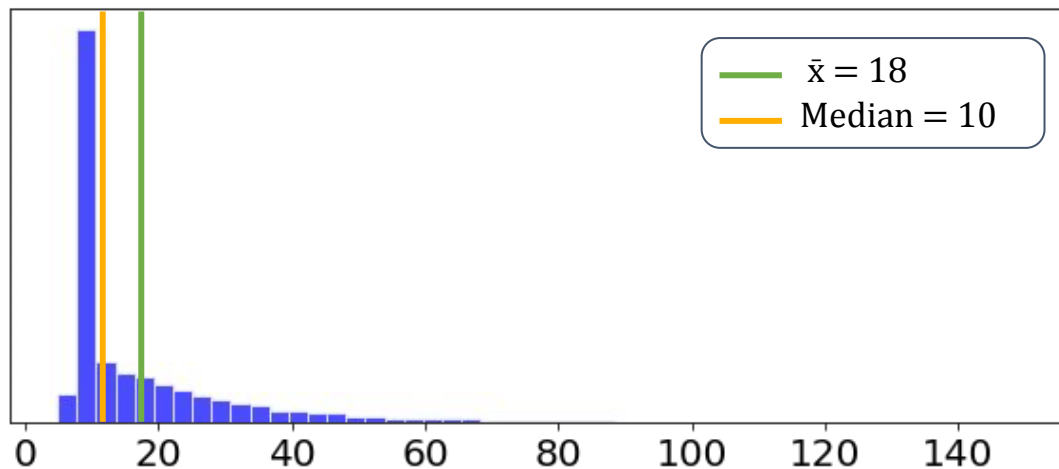
Hint: Consider the direction the mean is being pulled

Skewed positive

Skewed negative

Not skewed

Quick test!



Skewed positive

Skewed negative

Not skewed

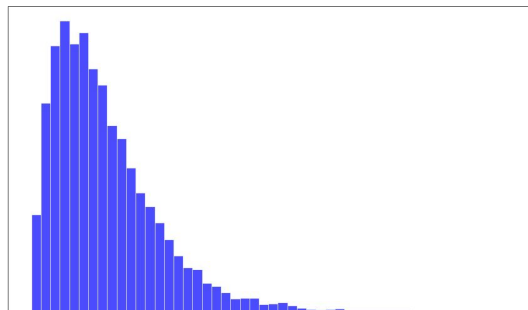
Skew in a spreadsheet



SKEW

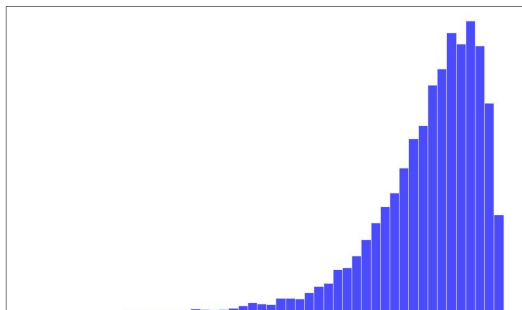
More positive values:

Positive skew



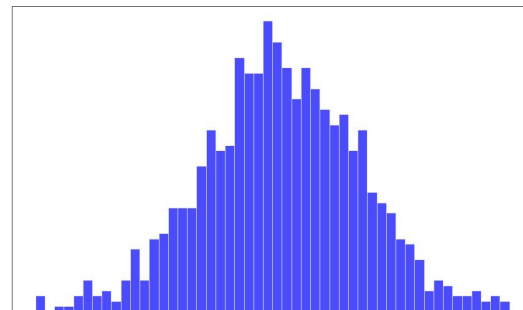
More negative values:

Negative skew



Values close to zero:

Minimal skew



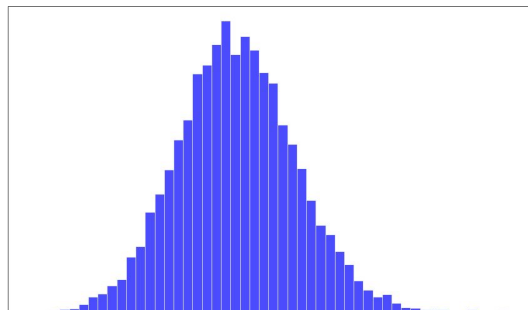
Skew in a spreadsheet



SKEW

Minimal skewness:

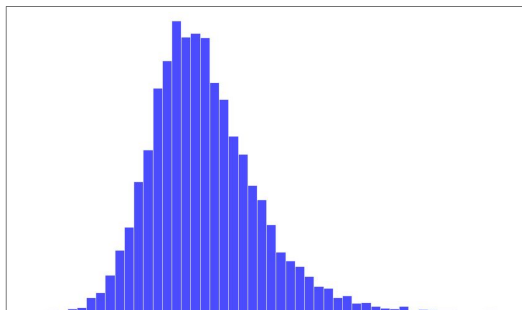
$$0 < |\text{skewness}| < 0.5$$



skewness = 0.17

Moderate skewness:

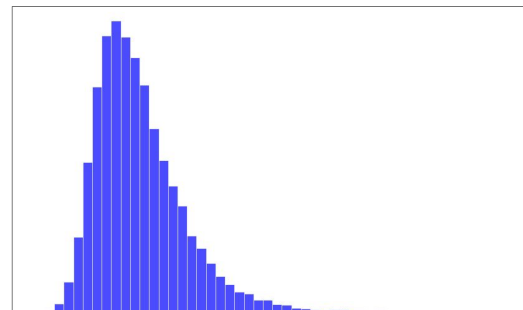
$$0.5 < |\text{skewness}| < 1$$



skewness = 0.78

High skewness:

$$|\text{skewness}| > 1$$



skewness = 1.22

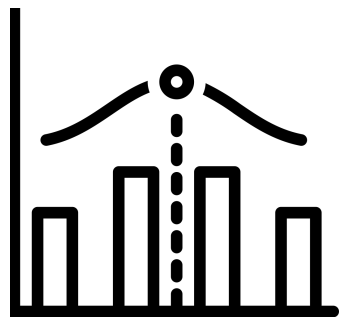


Foundational statistical techniques

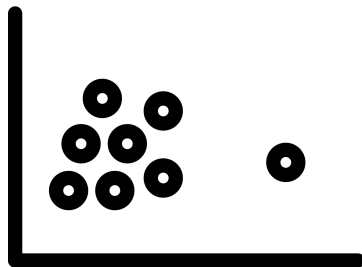
Why use these measures?

Why use these measures?

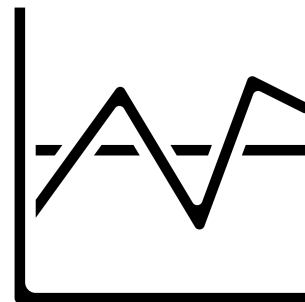
Depending on your business question, you may care about:



Average



Extremes



Variability

Scenario:

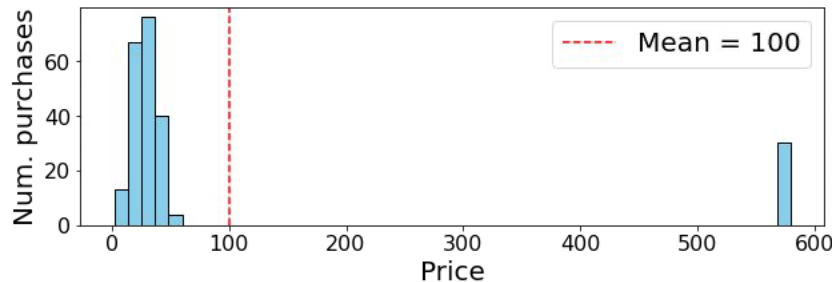


You
Data Analyst



Problem: "How much do people **typically** spend in one purchase?"

Central tendency

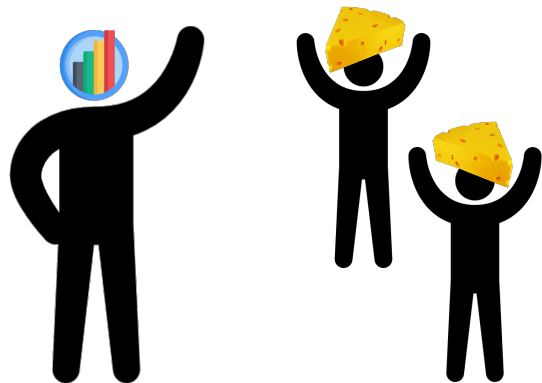


Mean of \$100 does not represent **typical** sale very well



Use **median** and investigate larger sales to derive insights on big spenders

Scenario:



You
Data Analyst

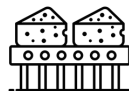
Variability



Problem: "How **consistent** is productivity across factories?"



Target:



Each factory to produce 50 cheese hats per day



Average productivity is 50 units per day

Confident forecast 👍

50 units

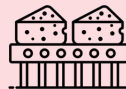


50 units

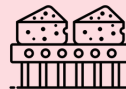


Different conclusion 👎

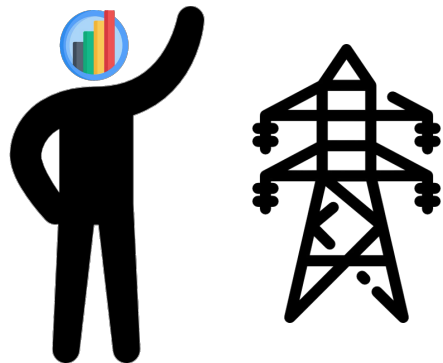
10 units



75 units



Scenario:



You
Data Analyst

Extremes



Problem: How much electricity capacity do we need to handle **peak** usage times?"



Look at **95th to 99th percentile**, to ensure you can handle peak load usage



Hot summer days



Cold winter nights



Foundational statistical techniques

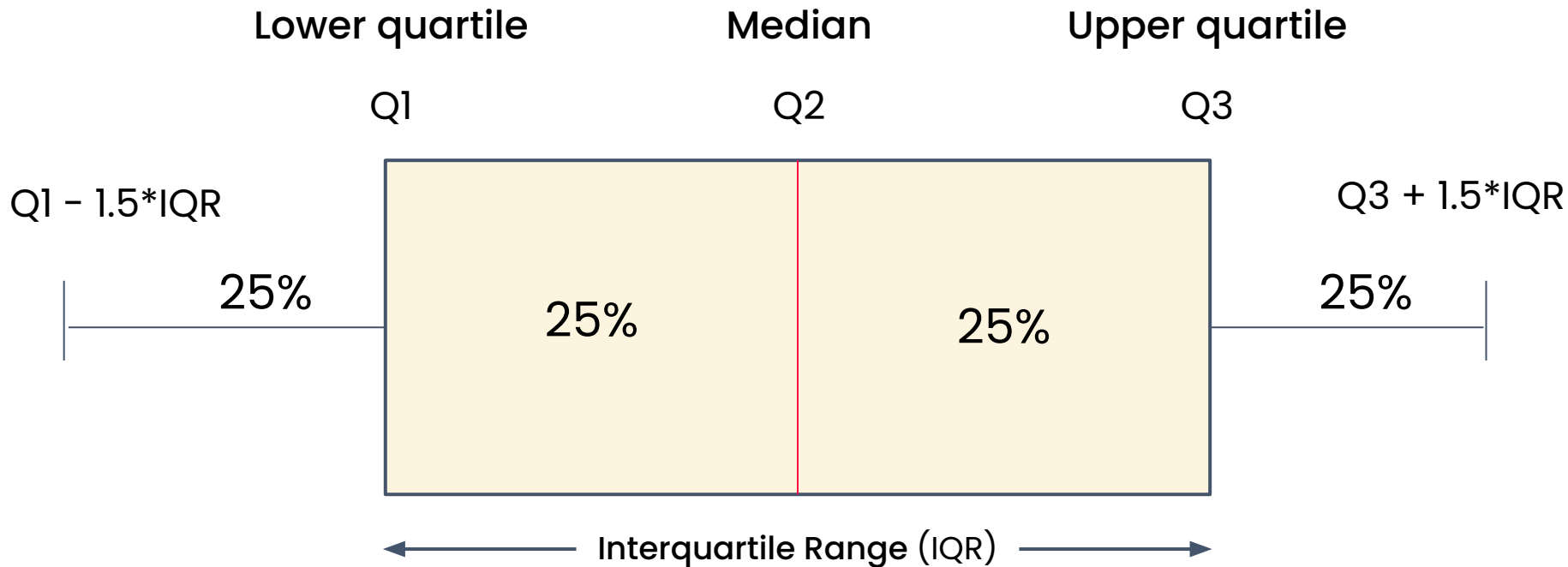
Demo: variability
and skewness



Foundational statistical techniques

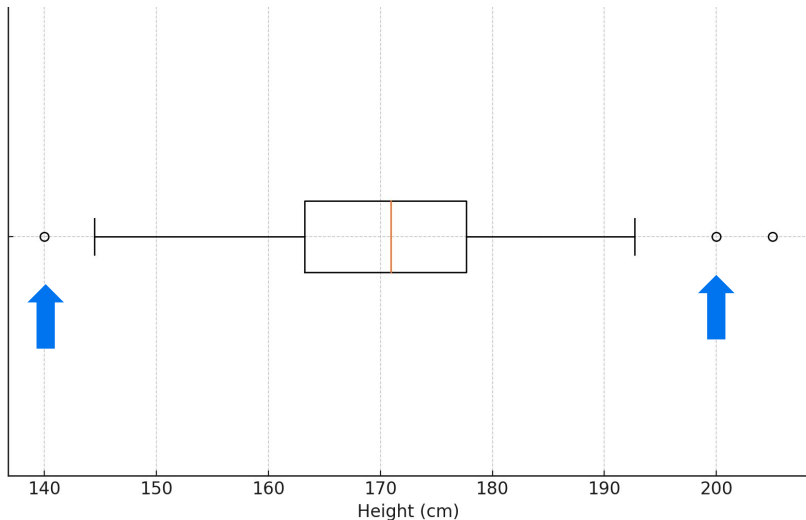
Box plots

What is a box plot?



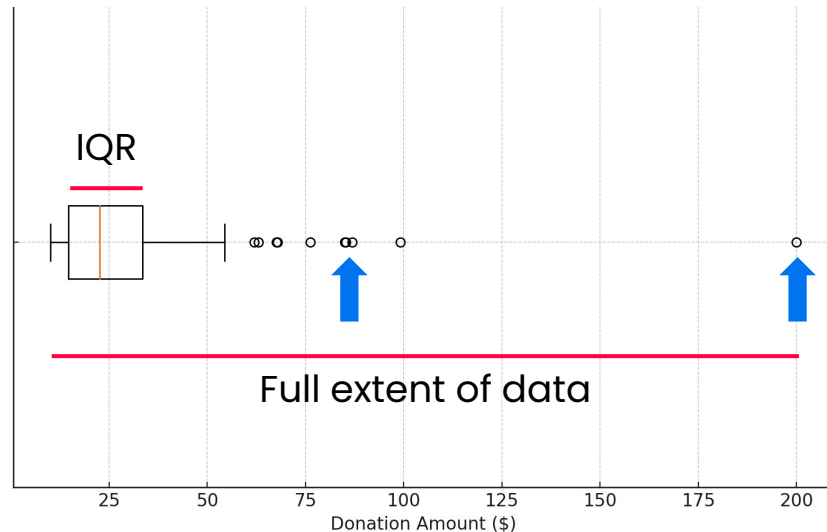
Outliers

Non-skewed distribution



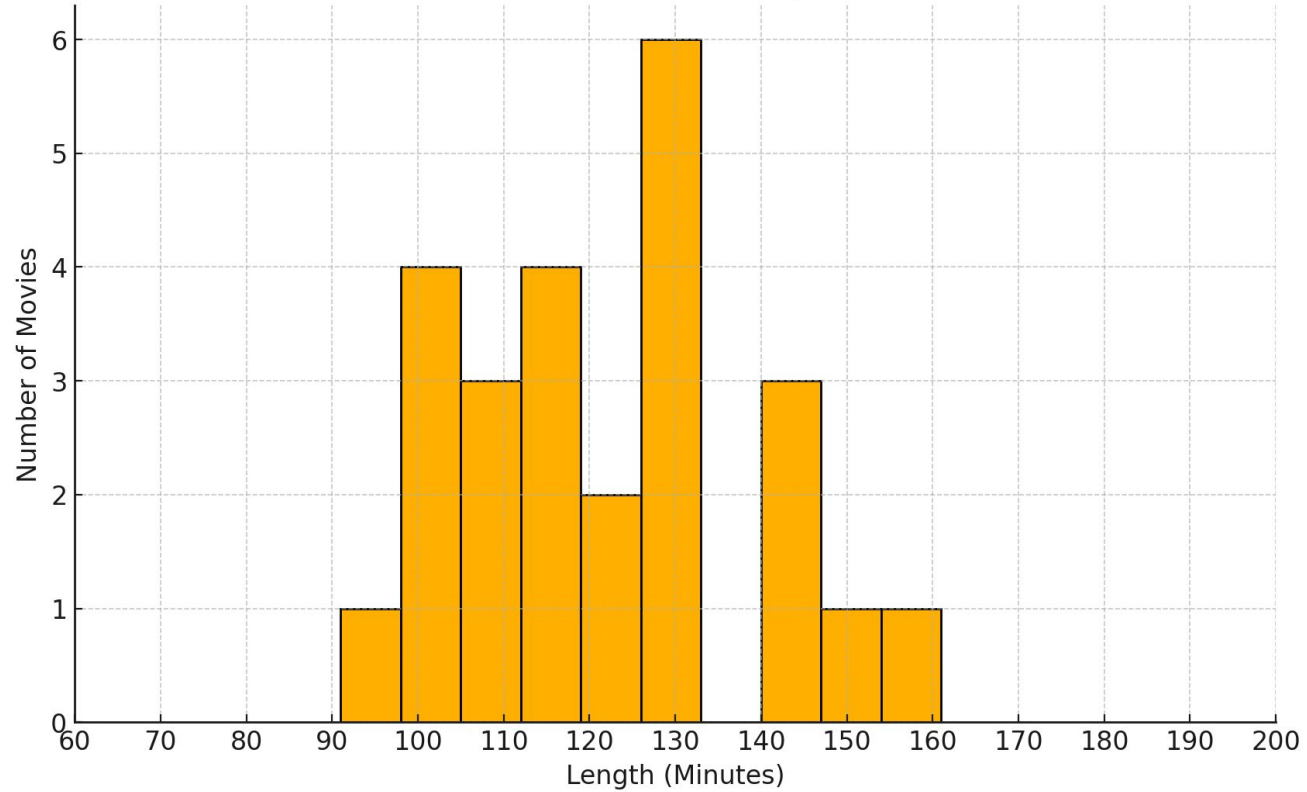
- Outliers could fall on **either side** of the distribution

Skewed distribution

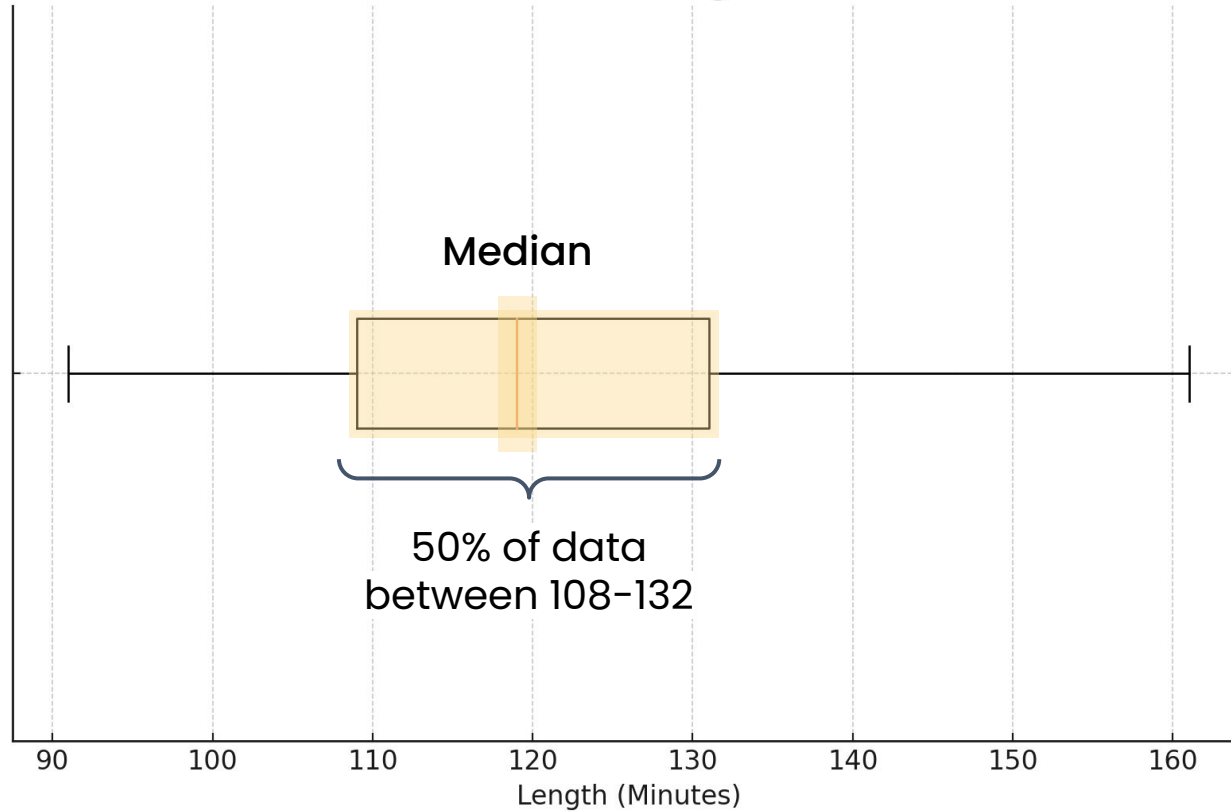


- Outliers are likely to fall on **one side** of the distribution

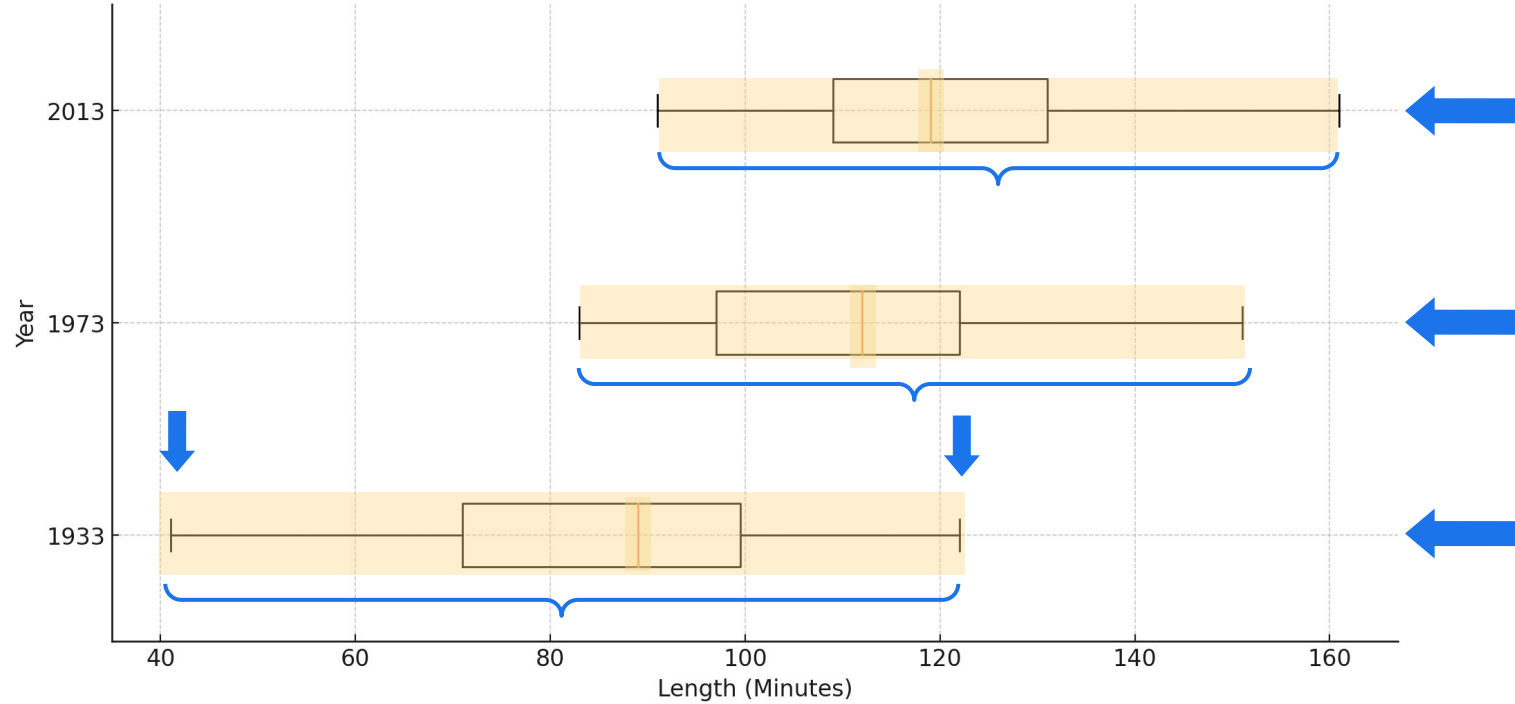
Distribution of Movie Lengths in 2013



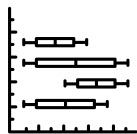
Box Plot of Movie Lengths in 2013



Box Plot of Movie Lengths in 1933, 1973, and 2013

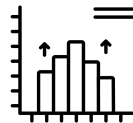


Box plot vs histogram



Boxplot

- Compare distributions across multiple groups
- Quickly identify outliers
- Display key summary statistics like **median**, **quartiles**, and **range**



Histogram

- Show detailed shape of your distribution
- Highlight frequencies of different value ranges
- Examine your data in granular detail



Foundational statistical techniques

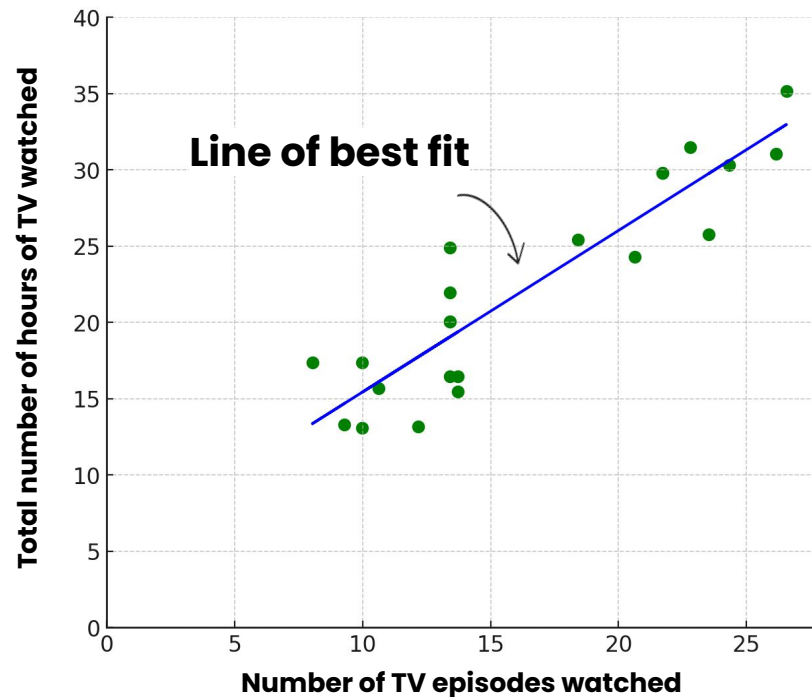
Demo: LLMs for spreadsheet
formulas & errors

Foundational statistical techniques

Correlation

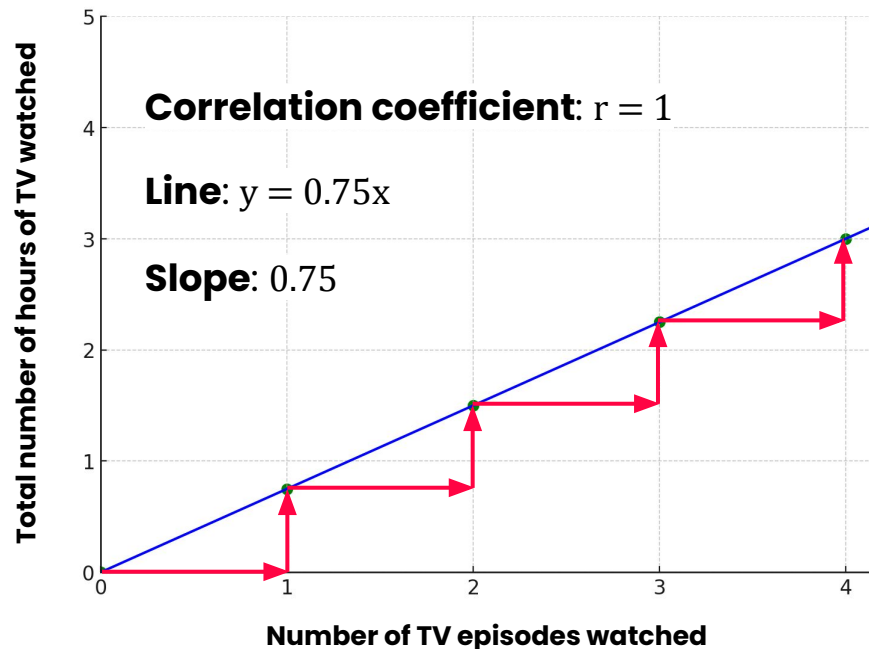
How to measure correlation

- Most common: **Pearson correlation coefficient**
 - Represented by r
 - Number between -1 and 1
- Measures the **strength** and **direction** of a **linear** relationship between two features



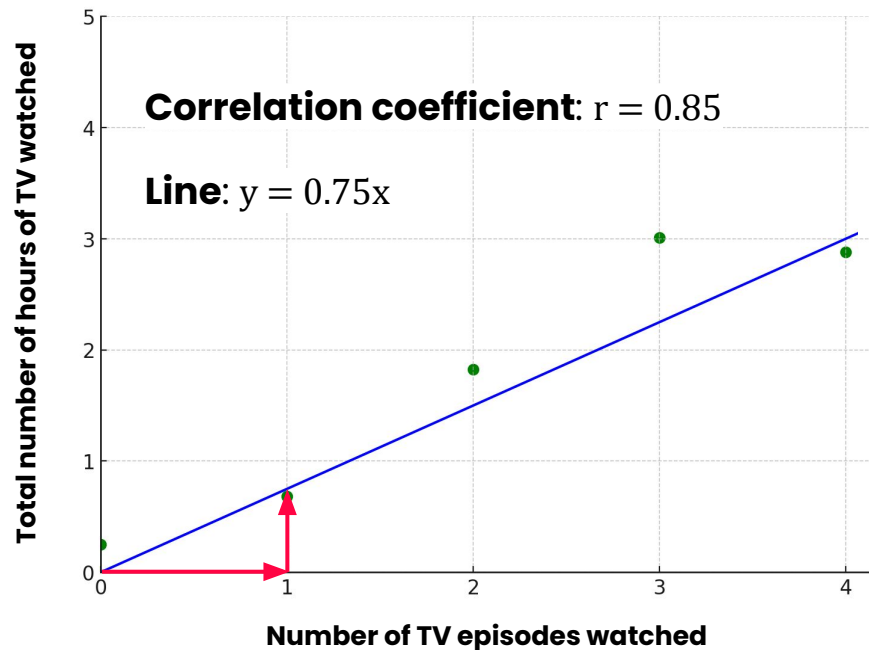
Example: perfect case

- A straight line has a consistent slope
 - For every unit change in X, Y changes by a consistent amount
 - **Example:** each TV show = 45 mins
- X and Y are **perfectly correlated**



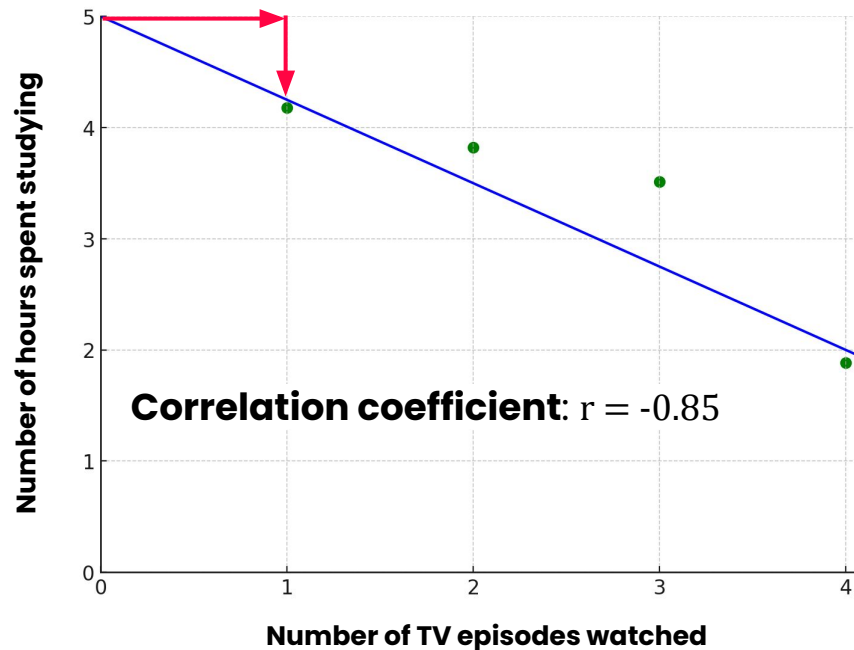
Example: less perfect case

- Each point is either a little above or a little below the line
- X goes up, Y goes up
- Episodes aren't the same length
- Line of best fit is the same, but correlation isn't as strong



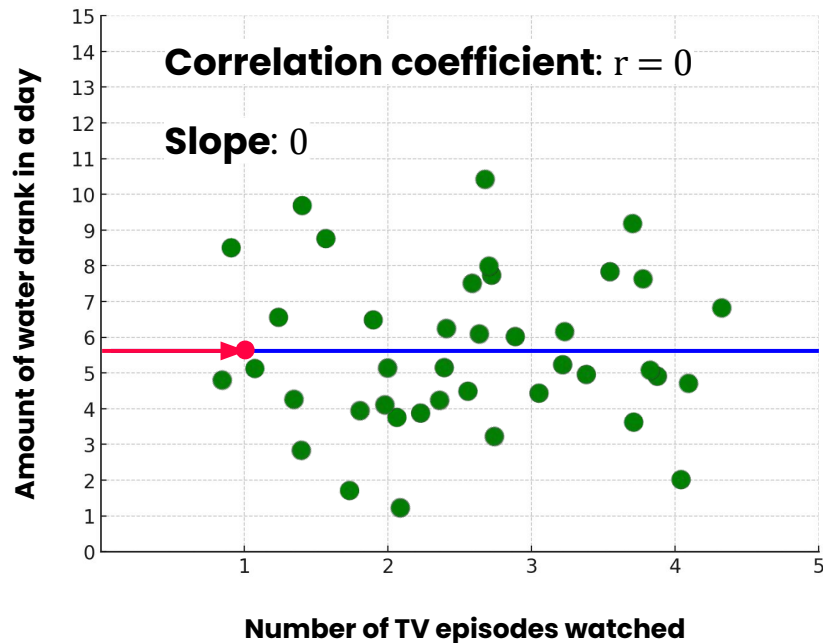
Example: negative correlation

- X goes up, Y goes down
- Line of best fit has a negative slope
- Correlation is equally as strong as the previous example
- Correlation is called “negative”



No correlation

- Line of best fit has a slope of 0
- X goes up, Y doesn't go up or down in any predictable way
- Line of best fit just predicts the mean value of Y



How to interpret r

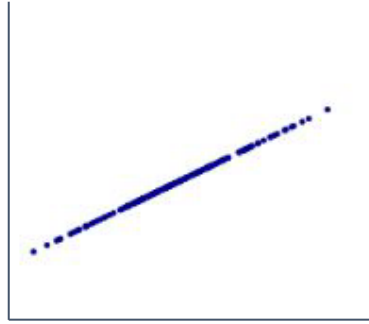
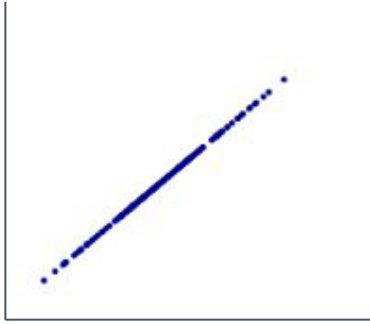
① Strength of the correlation

Absolute value of r	Strength
0 – 0.3	Weak
0.3 – 0.7	Moderate
0.7 – 1	Strong

② Direction of the correlation

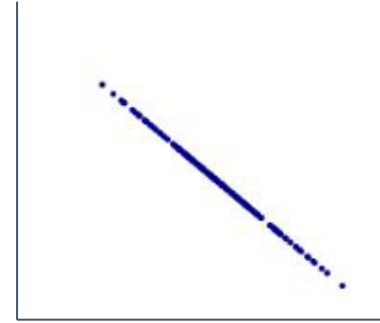
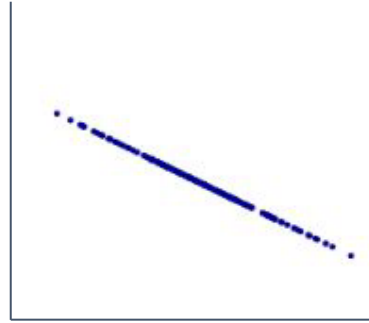
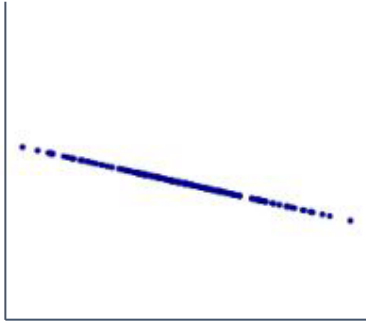
- **Positive:** both features increase or decrease together
- **Negative:** one feature increases, the other decrease

What do you think the r is for each?



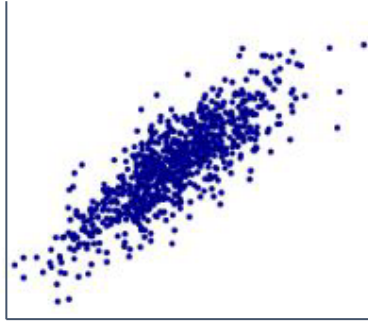
$r = 1$

What do you think the r is for each?

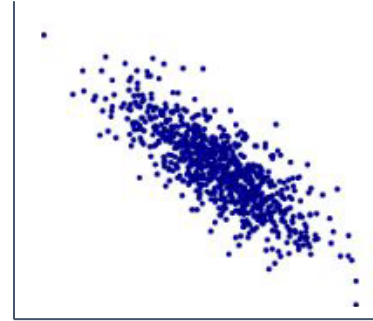


$$r = -1$$

What do you think the r is for each?

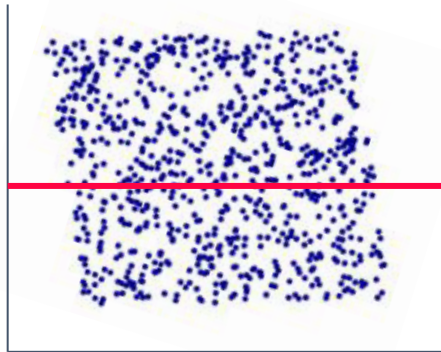


$r = 0.8$



$r = -0.8$

What do you think r is for this relationship?



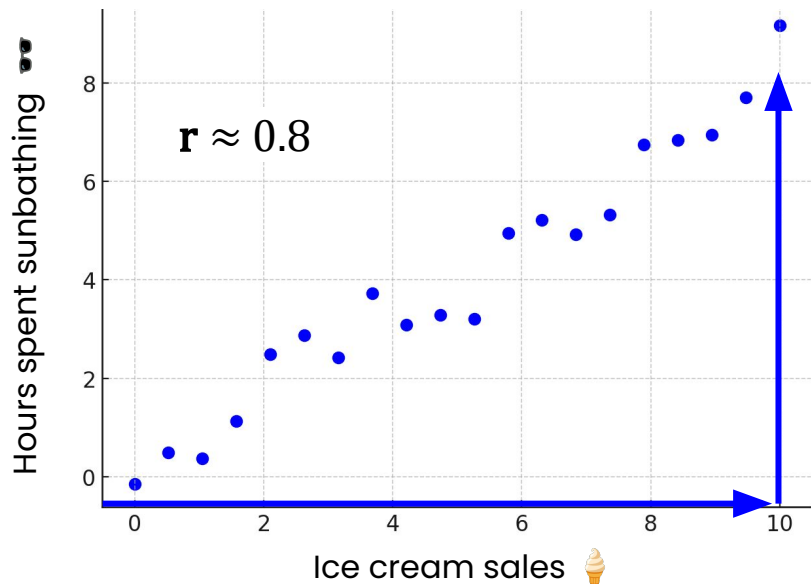
$$r = 0$$



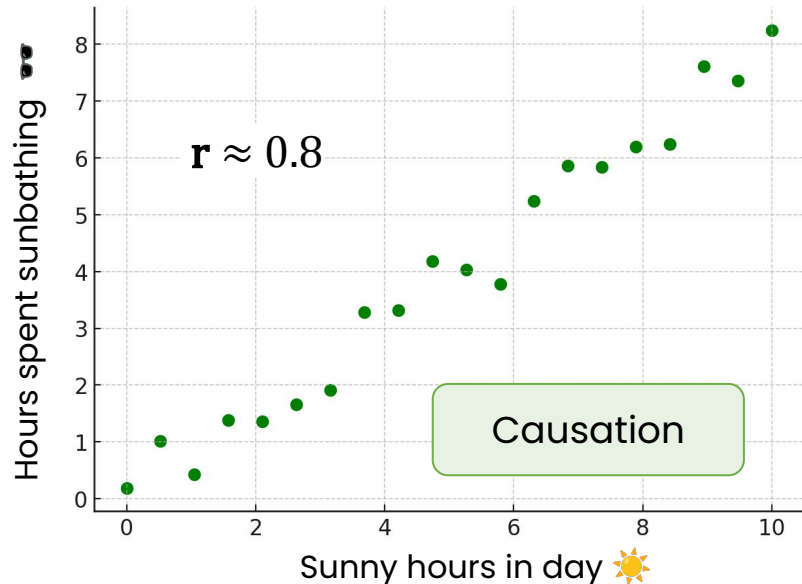
Foundational statistical techniques

Correlation and causation

Correlation and causation



- Although these two occurrences are correlated, one does not cause the other.



- ✓ One event is the result of another
- ✓ There's a cause-and-effect relationship

Correlation and causation

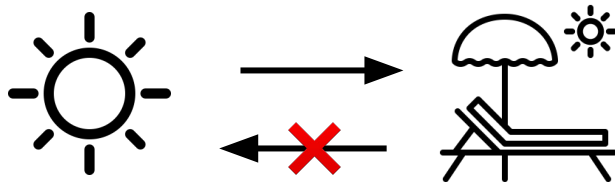
Correlation

- Indicates a relationship or association between two variables
- Does not imply that one feature causes the other
- Positive: move in the same direction
- Negative: move in opposite direction

- Third feature influencing both features
- Directionality problem

Causation

- Implies one feature directly affects the other
- Establishes a cause-and-effect relationship
- Requires more rigorous testing





Foundational statistical techniques

Demo: correlations &
scatterplots in spreadsheets



Foundational statistical techniques

What is segmentation?

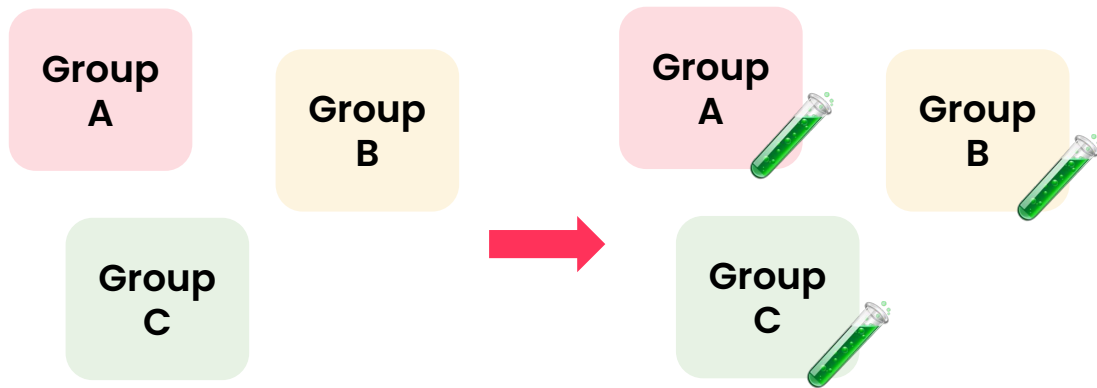
What is segmentation?

Dividing data into meaningful groups and analyzing each separately

- Analyzing feature or outcome across groups
 - Make different decisions based on characteristics
-  Age group
 Region
 Habits

Company	Segment by	In order to
Streaming service	Viewing habits	Tailor content recommendations
Healthcare provider	Risk factors	Determine interventions

Segmentation approach



Separate your data
into groups

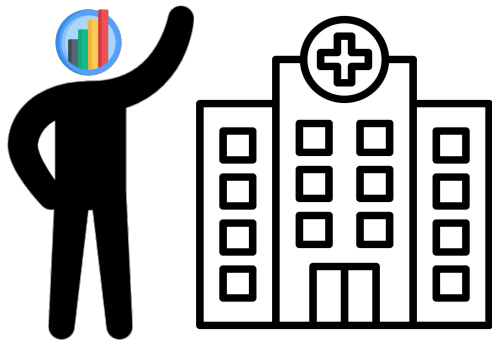
Run a **consistent** set
of analyses across
each segment

Set of analyses can include:

- ☐ Central tendency
- ☐ Variability
- ☐ Skew
- ☐ Correlation

☒ Same set of analyses:
able to compare results
across segments

Example



You
Data Analyst



Problem: Testing whether walking after surgery speeds recovery.



Median improvement in recovery time is **11%**

Median:

18-24
year olds



10.3%

25-44
year olds



11.1%

45-64
year olds



10.9%

Just looking at **overall median:**



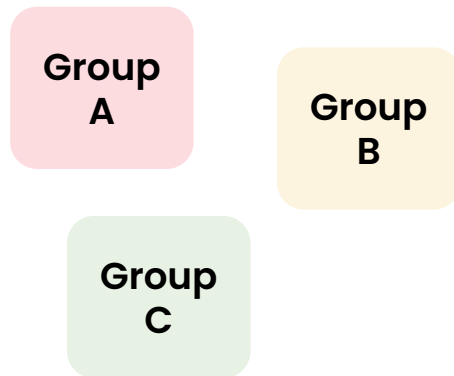
Equally valuable for everyone



Valuable for the average person

Understanding segmentation

- Doesn't have to have a formula
- Describes an approach of:
 - Breaking down the data
 - Trying to learn how subsets compare to each other



Visualizing segmented data

- **Bar and column charts**

Show how a feature is distributed across different categories

- **Grouped or stacked bar charts**

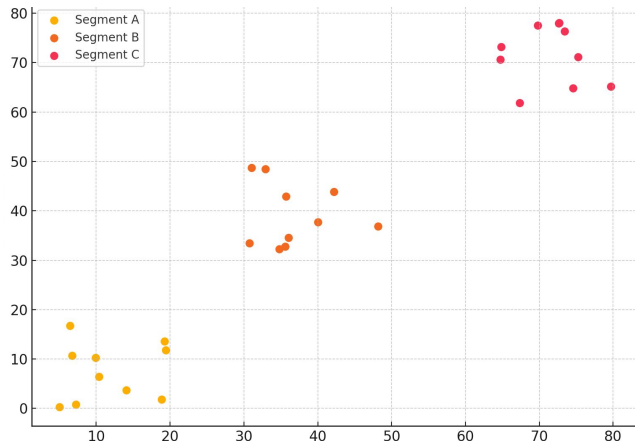
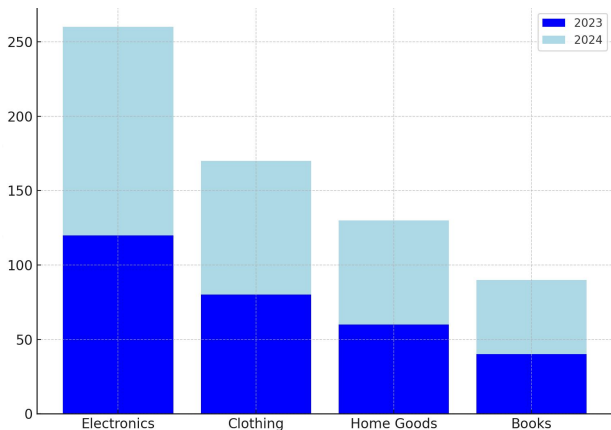
Two layers of segmentation

- **Scatter plot**

Color coding to show segments

- **Line charts**

Using multiple lines, each representing a group





Foundational statistical techniques

Demo: xlookup

Merging datasets



XLOOKUP

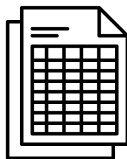
Combine data from multiple tables



Demographic data



Loan data



One customer

Many loans



Loan 1



Loan 2



Less efficient to store repeated customer information with each loan



Loan data

loan_id	customer_id
1	07113
2	91394



Key: unique identifier to access another table of data

ID of customer that took loan



Customer data

customer_id	name
80177	Pam
91394	Arnav



Customer who took loan



Foundational statistical techniques

Demo: pivot tables